



A7672X Series Hardware Design

LTE Module

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Document Title:	A7672X Series Hardware Design
Version:	V1.05
Date:	2023-08-16
Status:	Released

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Version History

Date	Version	Description of change	Author
2021-03-30	1.00	Initial	Zhongyou.chen Xuefeng.liu
2021-11-10	1.01	Current consumption data update.	Yu.xia Junxi Liu
2022-11-03	1.02	GNSS_PWRCTL line adds a series 10K resistor	Hao.li
2022-11-23	1.03	Add the section of reel packaging information	Hao.li
2023-02-13	1.04	1. Updated Table 1, Table 2, Table 30, Table 31, and Table 33: module GSM frequency band; 2. Add Table 46: GNSS current consumption 3. Updated the passive antenna reference	Hao.li
2023-08-16	1.05	Add GNSS Quick Start Indicator; Add GNSS_VBKP cannot be suspended description; Increase VBAT power-off and power-on restart timing; Add Bluetooth performance table; Recommended TVS model for adding antenna ports; Increase GNSS flow consumption; Modify ESD performance parameter table; Modify typical welding furnace temperature curve; Modify the description of humidity sensitive characteristics section.	Yang. Zhao

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1 Introduction

This document describes the hardware interface of the module, which can help users quickly understand the interface definition, electrical performance and structure size of the module. Combined with this document and other application documents, users can understand and use module to design and develop applications quickly. SIMCom provides a set of evaluation boards to facilitate module testing and use. The evaluation board tools include an EVB board, a USB cable, an antenna, and other peripherals.

1.1 Document Overview

The technical information of the module is mainly contained in the documents listed in Table 1, and all the documents in the following table must be read to fully understand the device and its applications.

Table 1: Overview of the A7672X documentation

No.	File Name	Description
1	A7672X Series Hardware Design_V1.05(this document)	It mainly introduces the interface function, recommended circuit, PCB layout guide, packaging and other hardware components, as well as the use of some AT instructions
2	SIMCom_A7670X&A7672X_Series_Reference_Design_20230818	Module reference circuit design
3	8BE000-A7678-TE_V2.01_SCH&PCB	TE schematic and PCB data
4	SIMCOM EVB Kit User Guide	EVB Kit user manual
5	A7670 &SIM800 &A7672 Series Compatible Design_V1.02	A7670&SIM800&A7672 series module difference comparison and compatible design
6	MOD_A7670C_R2_124	Module reference package
7	A76XX Series_AT Command Manual	AT instruction manual
8	Module Secondary SMT Process User Guide	This paper mainly introduces the SMT process of module production, the requirements of steel mesh production, reflow welding, welding and repair
9	A76XX series CAT1 module schematic diagram and PCB design checklist_V1.00	Module peripheral circuit schematic and PCB checklist

1.2 Product Outline

Aimed at the global market, the module supports GSM, LTE-TDD and LTE-FDD. Users can choose the module according to the wireless network configuration. The supported radio frequency bands are described in the following table.

Table 2: Module frequency bands

Standard	Frequency	A7672S	A7672E	A7672SA	A7672G
GSM	GSM850			✓	✓
	EGSM900	✓	✓	✓	✓
	DCS1800	✓	✓	✓	✓
	PCS1900			✓	✓
LTE-FDD	LTE-FDD B1	✓	✓	✓	✓
	LTE-FDD B2			✓	✓
	LTE-FDD B3	✓	✓	✓	✓
	LTE-FDD B4			✓	✓
	LTE-FDD B5	✓	✓	✓	✓
	LTE-FDD B7		✓	✓	✓
	LTE-FDD B8	✓	✓	✓	✓
	LTE-FDD B12				✓
	LTE-FDD B13				✓
	LTE-FDD B18				✓
	LTE-FDD B19				✓
	LTE-FDD B20		✓		✓
	LTE-FDD B25				✓
	LTE-FDD B26				✓
	LTE-FDD B28		✓*	✓	✓
	LTE-FDD B66			✓	✓
LTE-TDD	LTE TDD B34	✓			
	LTE TDD B38	✓			✓
	LTE TDD B39	✓			✓
	LTE TDD B40	✓			✓
	LTE TDD B41	✓			✓
Category		CAT1	CAT1	CAT1	CAT1
GNSS		Optional	Optional	Optional	
BlueTooth		Optional	Optional	Optional	

NOTE

- The functions supported by the module are based on the actual hardware model and software version.
- '*' indicate that only A7672E-LBSE in A7672E series supports B28, other models in A7672E series do not support B28.

With a small physical dimension of 24*24*2.4mm and with the functions integrated, the module can meet

almost any space requirement in users' applications, such as smart phone, PDA, industrial handheld, machine-to-machine and vehicle application, etc.

Module provides 124 pins, including 80 LCC pins in the outer ring and 44 LGA pins in the inner ring. This document will introduce all the functional pins.

1.3 Hardware Interface Overview

The interfaces are described in detail in the next chapters include:

- Power Supply
- USB 2.0 Interface
- Three UART Interface, one full function serial port, one ordinary serial port and one debug serial port
- Two USIM Interface
- One general ADC Interface
- One VBAT ADC Interface
- 4*4 matrix keyboard
- Analog audio input interface
- Analog audio output interface
- SPI Interface
- Two LDO Power Output
- One I2C Interface
- Multiple general input and output interfaces (GPIO)
- One LCD Interface
- One camera Interface
- One USB_BOOT interface
- Network status indication interface
- Module operation status indication interface
- Three antenna interface (GNSS and Bluetooth optional)

1.4 Hardware Block Diagram

The block diagram of the Module is shown in the figure below.

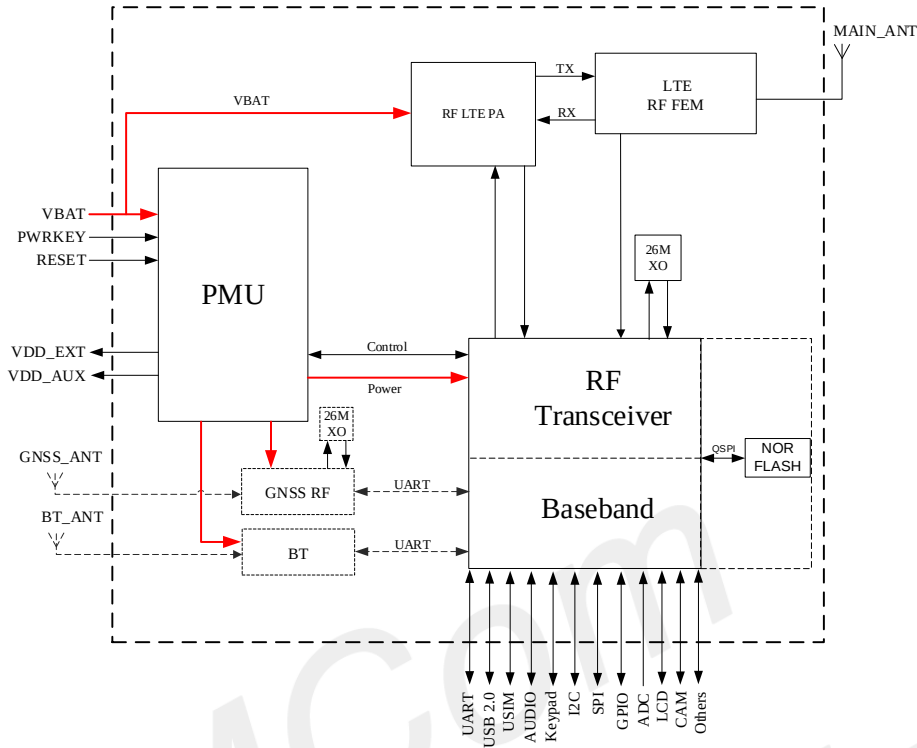


Figure 1: Module block diagram

1.5 Functional Overview

Table 3: General features

Feature	Implementation
Power supply	VBAT: 3.4V ~4.2V, Recommended VBAT: 3.8V
Power saving	Current in sleep mode: <2mA @AT+CSCLK=1 (VBAT=3.8V)
Radio frequency bands	Please refer to the table 2
Transmitting power	GSM/GPRS power level: -- GSM850/EGSM900: 4 (33dBm±2dB) -- DCS1800/PCS1900: 1 (30dBm±2dB) EDGE power level: -- GSM850/EGSM900: E2 (27dBm±3dB) -- DCS1800/PCS1900: E1 (26dBm+3dB/-4dB) LTE power level: 3 (23dBm±2.7dB)
Data Transmission Throughput	GPRS Multiple time slot level 12 EDGE Multiple time slot level 12 TDD/FDD-LTE category 1: 10 Mbps (DL), 5 Mbps (UL)
Antenna	GSM/LTE antenna interface GNSS antenna interface(optional) Bluetooth antenna interface(optional)
SMS	MT, MO, CB, Text, PDU mode Short Message (SMS) storage device: USIM Card, CB does not

	support saving in SIM Card Support CS domain and PS domain SMS
USIM interface	Support identity card: 1.8V/ 3V
USIM application toolkit	Support SAT class 3, GSM 11.14 Release 99 Support USAT
Phonebook management	Support phonebook types: SM/FD/ON/AP/SDN
Audio feature	Support one analog audio input interface Support one analog audio output interface
UART interface	●Full function serial port Baud rate support from 300bps to 3686400bps The default baud rate is 115200bps AT command and data can be sent through serial port Support RTS/CTS Hardware flow control ●Debug serial port Support debug usage, the default baud rate is 115200bps ●UART3 serial port The GNSS serial port is used for transparent data transmission by default, and the baud rate is 115200bps
I2C	●One channel I2C interface ●Comply with I2C bus protocol specifications ●Does not support multi host mode
USB	USB 2.0 compliant, host mode not supported. This interface can be used for AT command sending, data transmission, software debugging and upgrading.
Firmware upgrade	Firmware and FOTA upgrade over USB interface
Physical characteristics	Size:24*24*2.4mm Weight:2.8±0.1g
Networking protocol	●CMUX/PPP/MQTT/FTP/TCP/UDP/FTPS/MQTTS/HTTP/HTTPS/HTTP/NTP/PING/SSL/FILE ●PAP and CHAP authentication in accordance with PPP * The supported network protocols depend on the software version
Temperature range	Normal operation temperature: -30°C to +75°C Extended operation temperature: -40°C to +85°C* Storage temperature -45°C to +90°C

NOTE

1. When operating within the normal operating temperature range, the relevant performance of the module meets the requirements of the 3GPP standard;
2. Module is able to make and receive voice calls, data calls, SMS and make GPRS/LTE traffic in -40°C~85°C. The performance will be reduced slightly from the 3GPP specifications if the temperature is outside the normal operating temperature range and still within the extreme operating temperature range.

2 Package Information

2.1 Pin Assignment Overview

The following figure is a high-level view of the pin assignment of the module.

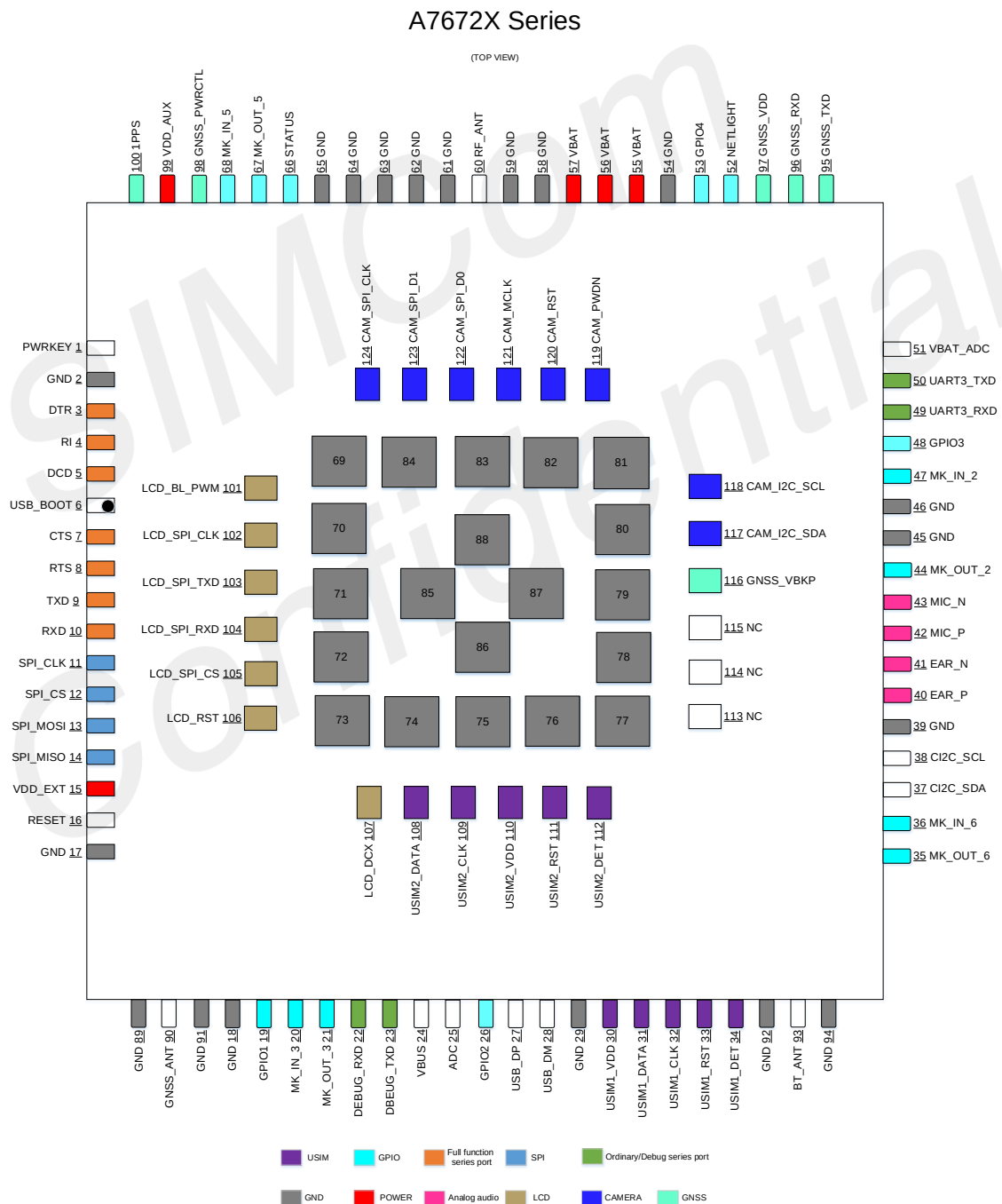


Table 4: Pin Description

Pin No	Pin Name	Pin No	Pin Name
1	PWRKEY	2	GND
3	DTR	4	RI
5	DCD	6	USB_BOOT●
7	CTS	8	RTS
9	TXD	10	RXD
11	SPI_CLK	12	SPI_CS
13	SPI_MOSI	14	SPI_MISO
15	VDD_EXT	16	RESET
17	GND	18	GND
19	GPIO1	20	MK_IN_3
21	MK_OUT_3	22	DEBUG_RXD
23	DEBUG_TXD	24	VBUS
25	ADC	26	GPIO2
27	USB_DP	28	USB_DM
29	GND	30	USIM1_VDD
31	USIM1_DATA	32	USIM1_CLK
33	USIM1_RST	34	USIM1_DET
35	MK_OUT_6	36	MK_IN_6
37	CI2C_SDA	38	CI2C_SCL
39	GND	40	EAR_P
41	EAR_N	42	MIC_P
43	MIC_N	44	MK_OUT_2
45	GND	46	GND
47	MK_IN_2	48	GPIO3
49	UART3_RXD	50	UART3_TXD
51	VBAT_ADC	52	NETLIGHT
53	GPIO4	54	GND
55	VBAT	56	VBAT
57	VBAT	58	GND
59	GND	60	RF_ANT
61	GND	62	GND
63	GND	64	GND
65	GND	66	STATUS
67	MK_OUT_5	68	MK_IN_5
69	GND	70	GND
71	GND	72	GND
73	GND	74	GND
75	GND	76	GND

77	GND	78	GND
79	GND	80	GND
81	GND	82	GND
83	GND	84	GND
85	GND	86	GND
87	GND	88	GND
89	GND	90	GNSS_ANT
91	GND	92	GND
93	BT_ANT	94	GND
95	GNSS_TXD	96	GNSS_RXD
97	GNSS_VDD	98	GNSS_PWRCTL
99	VDD_AUX	100	1PPS
101	LCD_BL_PWM	102	LCD_SPI_CLK
103	LCD_SPI_TXD	104	LCD_SPI_RXD
105	LCD_SPI_CS	106	LCD_RST
107	LCD_DCX	108	USIM2_DATA
109	USIM2_CLK	110	USIM2_VDD
111	USIM2_RST	112	USIM2_DET
113	NC	114	NC
115	NC	116	GNSS_VBKP
117	CAM_I2C_SDA	118	CAM_I2C_SCL
119	CAM_PWDN	120	CAM_RST
121	CAM_MCLK	122	CAM_SPI_D0
123	CAM_SPI_D1	124	CAM_SPI_CLK

NOTE

'●' Indicates that these Pins cannot be pulled down before the module powered up, otherwise it will affect the normal start-up of the module.

2.2 Pin Description

Table 5: Pin parameter abbreviation

Pin Type	Description
PI	Power input
PO	Power output
AI	Analog input

AO	Analog output
DIO	Digital input or output
DI	Digital input
DO	Digital output
DOH	Digital output with high level
DOL	Digital output with low level
PU	Pull up
PD	Pull down
OD	Open-drain

Table 6: 1.8V IO parameters definition

Power Domain	Parameter	Description	Min.	Typ.	Max.
1.8V	DC input conditions (VCC=1.8V)				
	V _{IH}	High level input	VCC * 0.7	1.8V	VCC + 0.2
	V _{IL}	Low level input	-0.3V	0V	VCC * 0.3
	R _{PU}	Pull up resistor	55KΩ	79 KΩ	121 KΩ
	R _{PD}	Pull down resistor	51 KΩ	87 KΩ	169 KΩ
	DC input conditions (VCC=1.8V Typical)				
	I _{IL}	Input leakage current	-	-	10uA
	DC output conditions (VCC=1.8V Typical)				
	V _{OH}	Output level range	VCC - 0.2	-	-
	V _{OL}	Output low range	-	-	0.2V
	I _{OL}	Maximum current driving capacity at low level output VPAD=0.2V IOL DCS[1:0]= 00 01 10 11	-	-	13 mA 25 mA 37 mA 49 mA
	I _{OH}	Maximum current driving capacity at high level output VPAD=VCC-0.2V IOH DCS[1:0]= 00 01 10 1	-	-	11 mA 21 mA 32 mA 42 mA

Table 7: 3.0V IO parameters definition

Power Domain	Parameter	Description	Min.	Typ.	Max.
3.0V	V _{IH}	High level input	2V	1.8V	VCC + 0.3
	V _{IL}	Low level input	-0.3V	0V	0.8V
	R _{PU}	Pull up resistor	26KΩ	47 KΩ	72 KΩ
	R _{PD}	Pull down resistor	27 KΩ	54 KΩ	267 KΩ
	I _{IL}	Input leakage current	-	-	10uA
	V _{OH}	Output level range	2.4V	-	-
	V _{OL}	Output low range	-	-	0.4V
	I _{OL}	Maximum current driving capacity at low level output VPAD=0.4V IOL DS[2:0]= 000	-	-	7 mA
		001			10 mA
		010			14 mA
		011			18 mA
		100			21 mA
		101			24 mA
		110			28 mA
		111			31 mA
	I _{OH}	Maximum current driving capacity at high level output VPAD=VCC-0.5V IOL DS[2:0]= 000	-	-	7 mA
		001			10 mA
		010			13 mA
		011			16 mA
		100			19 mA
		101			23 mA
		110			26 mA
		111			29 mA

Table 8: Pin description

Pin Name	Pin No.	Pin Parameter		Description	Note
		Power Domain	Type		
Power Supply					
VBAT	55,56, 57	-	PI	Module power input	Module input voltage ranges from 3.4V to 4.2V,

					Typical values is 3.8V.
VDD_EXT	15	-	PO	$V_{(nom)}=1.8V$ $IO_{(max)}=50mA$	Can provide power supply for GNSS. If unused, keep it open.
VDD_AUX	99	-	PO	$V_{(nom)}=3.0V$ $IO_{(max)}=50mA$	Can provide power supply for external device. If unused, keep it open.
GND	2,17,18, 29,39, 45,46, 54,58, 59,61, 62,63, 64,65, 69,70, 71,72, 73,74, 75,76, 77,78, 79,80, 81,82, 83,84, 85,86, 87,88, 89,91, 92,94	-	-	Ground	
System Control					
PWRKEY	1	-	DI,PU	Power ON/OFF input, active low. $V_{IH}: 0.7*VBAT$ $V_{IL}: 0.3*VBAT$ $V_{(nom)}=VBAT$	PWRKEY has been internally pulled-up to VBAT with 50KΩ resistor, default high.
RESET	16	-	DI,PU	System reset control input, active low. $V_{IH}: 0.7*VBAT$ $V_{IL}: 0.3*VBAT$ $V_{(nom)}=VBAT$	RESET has been pulled-up to VBAT with 50KΩ (typical) resistor, default high.
USIM Interface					
USIM1_DATA	31	1.8/3.0V	DIO,PU	USIM1 data	This pin has been pull-up with 4.7KΩ resistor to USIM1_VDD.
USIM1_RST	33	1.8/3.0V	DO,PU	USIM1 reset	
USIM1_CLK	32	1.8/3.0V	DO,PU	USIM1 clock	
USIM1_VDD	30	1.8/3.0V	PO	USIM1 card power supply	Module automatic recognition 1.8V or 3.0V USIM card.
USIM1_DET	34	1.8V	DI	USIM1 insert detect, it can be set to high/low active with the AT command, refer to	If unused, keep it open.

				Document [25]	
USIM2_DATA	108	1.8/3.0V	DIO,P U	USIM2 data	This pin need pull-up with 4.7KΩ resistor to USIM2_VDD externally.
USIM2_RST	111	1.8/3.0V	DO,PU	USIM2 reset	
USIM2_CLK	109	1.8/3.0V	DO,PU	USIM2 clock	
USIM2_VDD	110	1.8/3.0V	PO	USIM2 card power supply	Module automatic recognition 1.8V or 3.0V USIM card.
USIM2_DET	112	1.8V	DI,PD	USIM2 insert detect, it can be set to high/low active with the AT command, refer to Document [25]	If unused, keep it open.
USB Interface					
VBUS	24	-	PI	Valid USB detection input. Active high	V _{min} =3.0V, V _{max} =5.2V
USB_DM	28	-	AIO	Negative line of the differential, bi-directional USB signal.	90Ω differential impedance.
USB_DP	27	-	AIO	Positive line of the differential, bi-directional USB signal.	
Full Function UART Interface					
RTS	8	1.8V	DI	DTE request sending	Connect to the RTS of DTE, If unused, keep it open.
CTS	7	1.8V	DO	DTE clear sending	Connect to the CTS of DTE, If unused, keep it open.
RXD	10	1.8V	DI	Data input	If unused, keep it open.
TXD	9	1.8V	DO	Data output	
RI	4	1.8V	DO	Ring indicator	
DCD	5	1.8V	DO	Carrier detection	
DTR	3	1.8V	DI	DTE Ready	
Debug UART					
DEBUG_TXD	23	1.8V	DO	Debug UART output	Default used as debug port.
DEBUG_RXD	22	1.8V	DI	Debug UART input	
Serial Port UART3					
UART3_TXD	50	1.8V	DO	UART3 output	If unused, keep it open.
UART3_RXD	49	1.8V	DI	UART3 input	
I2C Interface					
CI2C_SCL	38	1.8V	OD	I2C clock output	If unused, keep it open. Need pull
CI2C_SDA	37	1.8V	OD	I2C data I/O	

					up to VDD_EXT externally.
SPI Interface					
SPI_CLK	11	1.8V	DO	SPI clock	If unused, keep it open.
SPI_CS	12	1.8V	DO	SPI chip selection	
SPI_MOSI	13	1.8V	DO	SPI master output slave input	
SPI_MISO	14	1.8V	DI	SPI master input slave output	
Analog Audio Interface					
EAR_P	40	1.8V	AO	Earphone output positive	If unused, keep it open.
EAR_N	41	1.8V	AO	Earphone output negative	
MIC_P	42	1.8V	AI	MIC input positive	
MIC_N	43	1.8V	AI	MIC input negative	
GPIO					
GPIO1	19	1.8V	DIO,PU	General purple I/O	If unused, keep it open.
GPIO2	26	1.8V	DIO,PD	General purple I/O	If unused, keep it open.
GPIO3	48	1.8V	DIO,PU	General purple I/O	If unused, keep it open.
GPIO4	53	1.8V	DIO,PU	General purple I/O	If unused, keep it open.
GNSS Interface					
GNSS_PWRCTL	98	1.8V	DI	The enable control PIN of GNSS power supply.	Active high.
GNSS_VDD	97	-	PI	The power input for GNSS, Voltage range not less than 1.8V, but not higher than 1.9V	Module VDD_EXT (PIN 15) can be used for this power supply.
GNSS_VBKP	116	-	PI	GNSS VRTC power input, input voltage 1.4V~3.6V	This pin cannot be floated.
1PPS	100	1.8V	DO	1PPS signal output	If unused, keep it open.
GNSS_RXD	96	1.8V	DI	GNSS UART RX	Connect to MCU UART_TX; Or use 1K resistors in series in module UART3_TX (pin 50).
GNSS_TXD	95	1.8V	DO	GNSS UART TX	Connect to MCU UART_RX; Or use 1K

					resistors in series in module UART3_RX (pin 49).
LCD Interface					
LCD_BL_PWM	101	1.8V	DO	LCD backlight adjusting PWM	If unused, keep it open.
LCD_SPI_CLK	102	1.8V	DO	SPI clock	
LCD_SPI_TXD	103	1.8V	DIO	SPI DATA(Bidirectional)	
LCD_SPI_RXD	104	1.8V	DI	SPI DATA	
LCD_SPI_CS	105	1.8V	DO	SPI CS	
LCD_RST	106	1.8V	DO	LCD Reset	
LCD_DCX	107	1.8V	DO	Command/parameter selection	
CAMERA Interface					
CAM_I2C_SDA	117	1.8V	OD	Camera I2C data	If unused, keep it open.
CAM_I2C_SCL	118	1.8V	OD	Camera I2C clock	
CAM_PWDN	119	1.8V	DO	Camera power down	
CAM_RST	120	1.8V	DO	Camera reset	
CAM_MCLK	121	1.8V	DO	Camera main clock	
CAM_SPI_D0	122	1.8V	DIO	Camera SPI D0	
CAM_SPI_D1	123	1.8V	DIO	Camera SPI D1	
CAM_SPI_CLK	124	1.8V	DO	Camera SPI clock	
ANT Interface					
RF_ANT	60	-	AIO	Main antenna	
GNSS_ANT	90	-	AIO	GNSS antenna	
BT_ANT	93	-	AIO	Bluetooth antenna	
Matrix keyboard interface					
MK_IN_2	47	1.8V	DI	Matrix keyboard input	If unused, keep it open.
MK_IN_3	20	1.8V	DI	Matrix keyboard input	
MK_IN_5	68	1.8V	DI	Matrix keyboard input	
MK_IN_6	36	1.8V	DI	Matrix keyboard input	
MK_OUT_2	44	1.8V	DO	Matrix keyboard output	
MK_OUT_3	21	1.8V	DO	Matrix keyboard output	
MK_OUT_5	67	1.8V	DO	Matrix keyboard output	
MK_OUT_6	35	1.8V	DO	Matrix keyboard output	
Other Pins					
ADC	25	-	AI	General Purpose analog-to-digital converter interface	If unused, keep it open. Input voltage ranges from 0 to 1.8V.
VBAT_ADC	51	-	AI	VBAT analog-to-digital	If unused, keep it

				converter interface	open.
NETLIGHT	52	1.8V	DO	Network registration status indicator (LED).	If unused, keep it open.
STATUS	66	1.8V	DO	Module status indicator (LED). High: power on Low: power off	If unused, keep it open.
USB_BOOT●	6	1.8V	DI	Firmware download guide control input. When pull-up to GND and press PWRKEY, module will access in USB download mode.	Do place 2 test points for debug. Do not pull down USB_BOOT during normal power up!
NC	113, 114, 115	-	-	-	Keep open.

NOTE

1. Please reserve a test point for USB_BOOT, VDD_EXT and DEBUG_TXD. If there is no USB connector, please also reserve a test point for USB_VBUS, USB_DP, and USB_DM for Firmware upgrade.
2. '●' indicates that these signals cannot be pulled down before startup, otherwise it will affect the normal startup of the module.

2.3 Mechanical Information

The following figure shows the package outline drawing of MODULE.

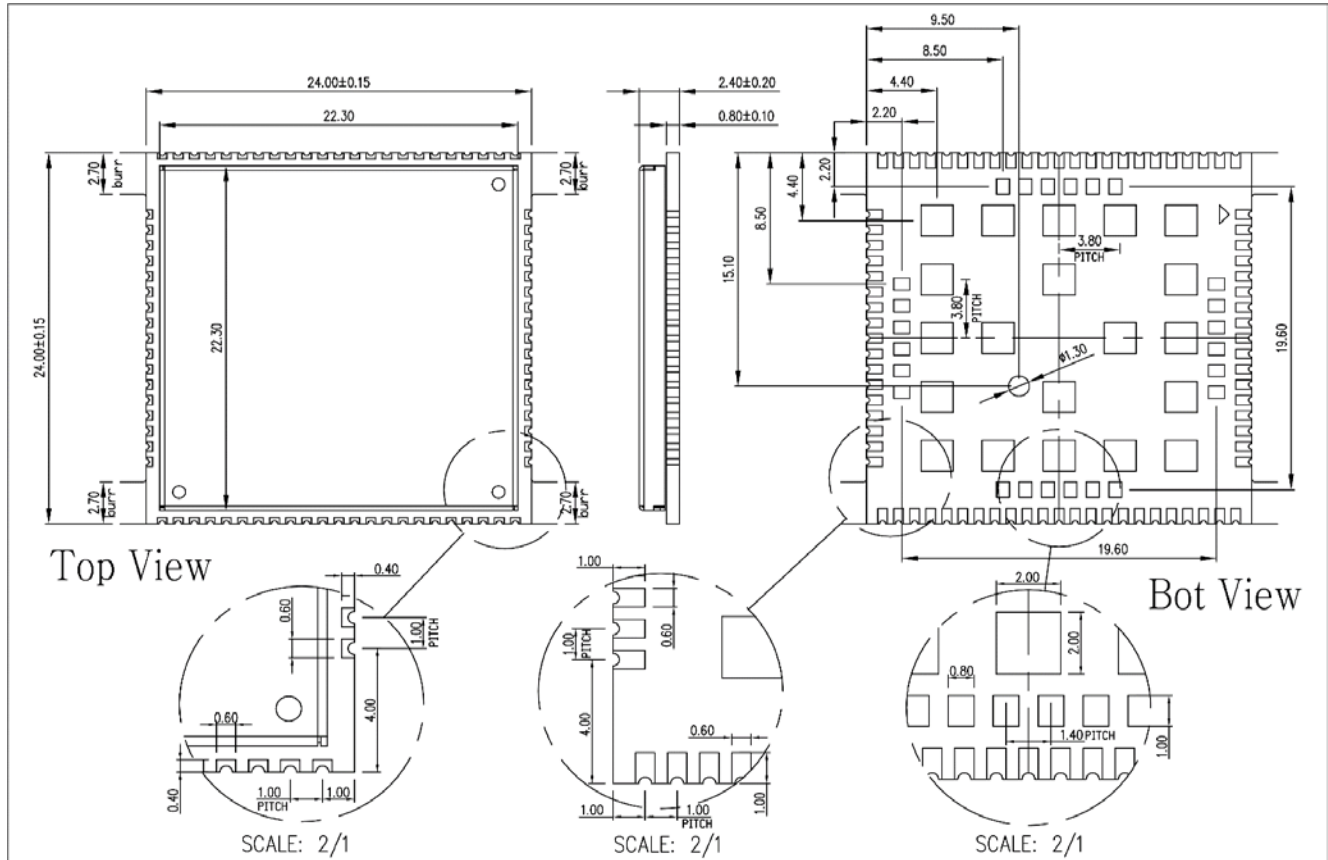


Figure 3: Dimensions (Unit: mm)

NOTE

The side length dimension is 24.00±0.15mm excluding the burr area.

Top View

Figure 4: Footprint recommendation (Unit: mm)

2.5 Recommend Stencil Size

Recommend stencil thickness $\geq 0.12\text{mm}$ and $< 0.15\text{mm}$.

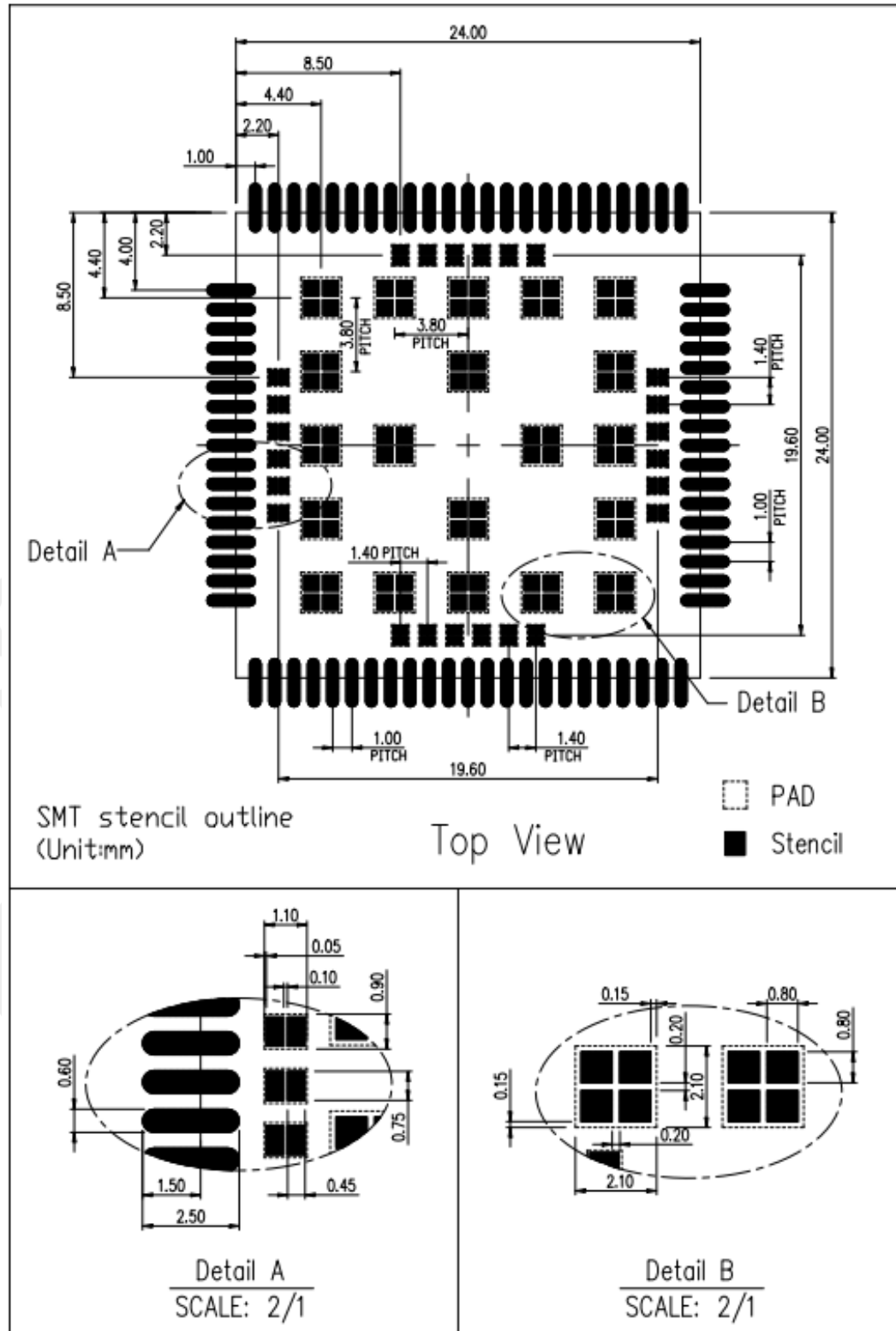


Figure 5: Recommend stencil dimension (Unit: mm)

3 Interface Application

3.1 Power Supply

Module uses a single power supply, with a total of 3 pins (55, 56, and 57) as VBAT power input pin. Module use these three pins supply the internal RF and baseband circuit.

When the module is at the maximum power in GSM TX mode, the peak current can reach 2A (peak current), which results in a large voltage drop on VBAT. In order to ensure that the voltage drop is less than 300mV, the power supply capacity of external power supply must be no less than 2A.

When the module is at the maximum power in LTE TX mode, the peak current can reach 700mA (peak current), which results in a large voltage drop on VBAT. In order to ensure that the voltage drop is less than 300mV, the power supply capacity of external power supply must be no less than 700mA.

Table 9: VBAT power supply interface

Pin Name	Pin No.	I/O	Description	Note
VBAT	55,56,57	PI	Module power supply input	The input voltage ranges from 3.4V to 4.2V. The external power supply must provide 2A current carrying capacity.
GND	2,17,18,29,39,45,46,54,58,59,61~65,69~89,91,92,94			

Table 10: VBAT pins electronic characteristic

Parameter	Description	Min.	Typ.	Max.	Unit
V _{BAT}	Module supply voltage	3.4	3.8	4.2	V
I _{VBAT} (GSM)	Module consumption peak current in GSM mode	-	2	-	A
I _{VBAT} (LTE)	Module consumption peak current in LTE mode	-	700	-	mA
I _{VBAT} (average)	Module average consumption current (normal mode)	-	21	-	mA
I _{VBAT} (sleep)	Module average consumption current (sleep mode)	-	2.5	-	mA
I _{VBAT} (power-off)	Module average consumption current (off leakage current)	-	20	-	uA

NOTE

- Test condition: VBAT power supply 3.8V, the module is tested on EVB board, and the power input has a 200uF tantalum capacitor.
- The I_{VBAT} data in the preceding table is the overall flow consumption data of the module.

3.1.1 Power Supply Design Guide

In the user's design, special attention must be paid to the design of the power supply. If the voltage drops below 3.4V, the RF performance of the module will be affected, the module will shut down if the voltage is too low. It is recommended to select an LDO or DC-DC chip with an enable pin, and the enable pin is controlled by the MCU.

NOTE

When the power supply can provide a peak current of 2A, the total capacity of the external power supply capacitance is recommended to be no less than 200uF, to ensure that the voltage drop on the VBAT pin at any time is not under 3.4V.

It is recommended to place four 10pF/33pF/0.1uF/1uF ceramic capacitors near VBAT to improve RF performance and system stability. At the same time, it is recommended that the VBAT layout routing width from the power supply on the PCB to the module be at least 2mm. In principle, the longer the VBAT line, the wider the line width.

If the VBAT input contains high-frequency interference, it is recommended to add magnetic beads for filtering. The recommended types of magnetic beads are BLM21PG300SN1D and MPZ2012S221A.

Reference design recommendations are as follows:

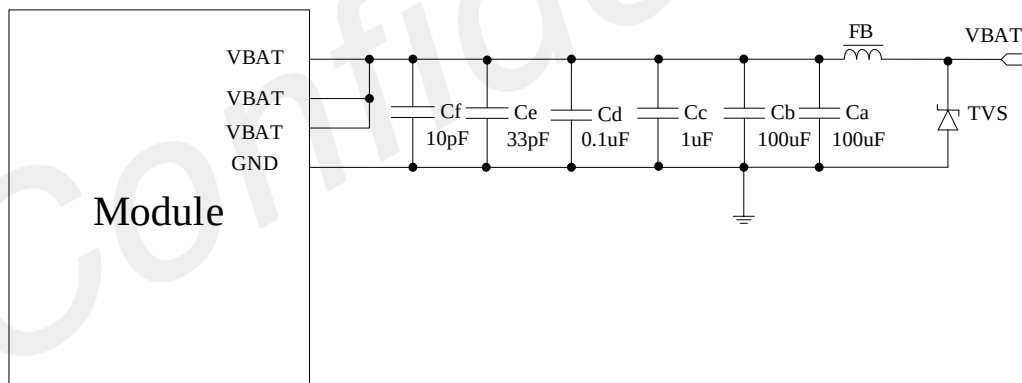


Figure 6: Power supply application circuit

In addition, in order to prevent the damage of module caused by surge and overvoltage, it is recommended to parallel one TVS on the VBAT pin of the module.

Table 11: Recommended TVS diode list

No.	Manufa cturer	Part Number	V_{RWM}	$V_C(max)$	$P_{pp}(max)$	$C_J(max)$	Package
-----	------------------	-------------	-----------	------------	---------------	------------	---------

1	WILL	ESD56301D05	5V	9.5V	1500W	700pF	DFN1610-2L
2	WILL	ESD56301D04-2/TR	4.85V	11V	2000W	480pF	DFN1610-2L
3	WAYON	WS2057KP	5V	12V	2040W	700pF	DFN1610-2L
4	WAYON	WS4.5DPHXM	4.85V	11V	2255W	700pF	DFN1610-2L

NOTE

When selecting TVS by customer, it is necessary to pay attention to the clamping voltage in the case of surge protection. The clamping voltage should not be higher than 10V when 100V surge input.

3.1.2 Recommended Power Supply Circuit

The MCU must have the function to power off the module, but the module cannot be shut down or restarted normally. Only when the module is abnormal and cannot be shut down or restarted normally can the module be powered off. When the input power is greater than 9V, the DCDC chip is recommended. When the input is less than 9V, it is recommended to use LDO power supply. If you use the module's OPEN LINUX secondary development function, because there is no MCU, you can add a low-cost single-chip microcomputer to play the role of hardware watchdog to pull POWERKEY to boot and can be powered off. The following figure shows the DC-DC regulator reference circuit:

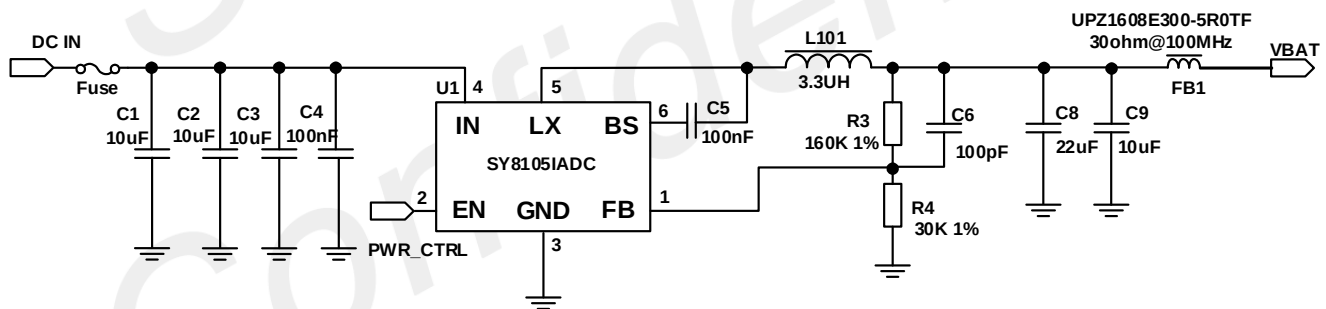


Figure 7: Power supply reference circuit

When the VBAT power is turned off, the voltage should decrease rapidly. To avoid voltage anomalies, when the VBAT is lower than the minimum value, it must be pulled below 100mV for at least 2 seconds before the system is powered up again.

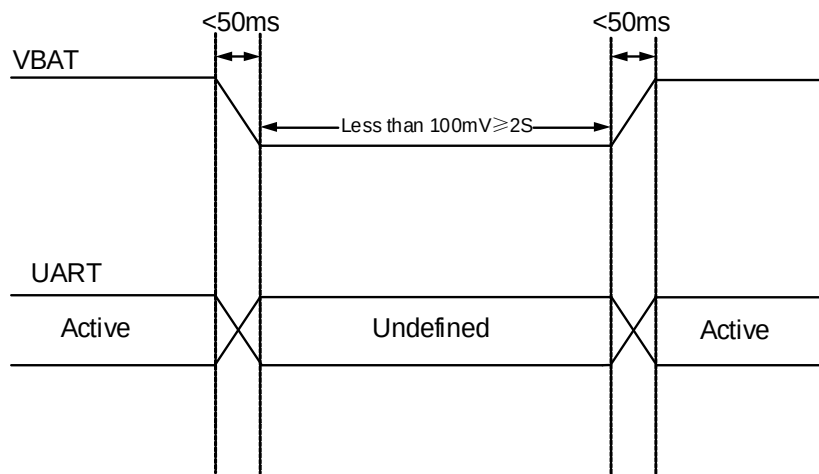


Figure 8: Power-off and power-on restart sequence

3.1.3 Voltage Monitor

AT command 'AT+CBC' can be used to monitor VBAT voltage.

AT command 'AT+CVALARM' can be used to set high/low voltage alarm, When the actual voltage exceeds the preset range, a warning message will be reported through the AT port.

AT command 'AT+CPMVT' can be used to set high/low voltage power off, When the actual voltage exceeds the preset range, the module will shut down automatically.

NOTE

Overvoltage alarm and overvoltage shutdown function are off by default. For details of AT commands, please refer to document [1].

3.2 Power On/Off and Reset

3.2.1 Power on

Table 12: PWRKEY interface

Pin Name	Pin No.	I/O	Description	Note
PWRKEY	1	DI	Module power on/off	Control the module power on/off.

Customer can power on the module by pulling down the PWRKEY pin. This pin has been pulled up to

VBAT inside the module.

It is recommended that when using the module, adding TVS diode at the module pin can effectively enhance the ESD performance.

The recommended circuit is as follows:

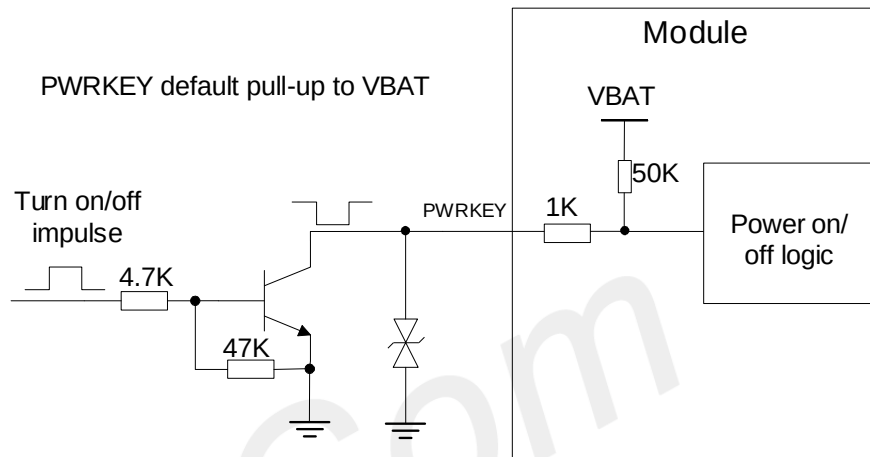


Figure 9: Reference power on/off circuit

NOTE

Do not parallel capacitors which the value is exceed 100nF on PWRKEY. It will cause module power on automatically when VBAT powered. At the same time, pay attention to the junction capacitance of TVS. If the junction capacitance is too large, there will be the same risk.

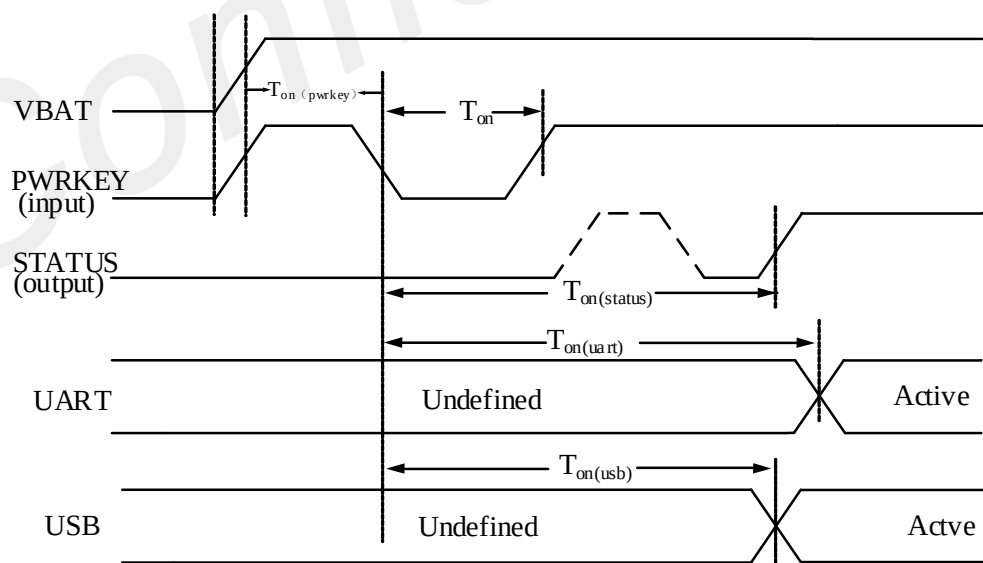


Figure 10: Power on timing sequence

NOTE

1. In earlier software versions, when the module did not have an external USB, the STATUS would briefly output a high-level pulse during the module startup process and then output a low-level signal. After the module initialization was completed, it would output a continuous high-level signal; When the module is externally connected to USB, there will be no short high-level pulses during the module startup process. Only after the module initialization is completed, the STATUS immediately outputs a continuous high-level pulse;
2. In the latest software version, STATUS is not affected by USB and only outputs a continuous high level once the module initialization is completed.

Table 13: Power on timing and electronic characteristic

Symbol	Parameter	Min.	Typ.	Max.	Unit
$T_{on(pwrkey)}$	PWRKEY pull-down time (The module can be started normally)	-	30	-	ms
T_{on}	The time of active low-level impulse of PWRKEY pin to power on module	-	50	-	ms
$T_{on(status)}$	The time from power-on issue to STATUS pin output high level (indicating power up ready)	-	7	-	s
$T_{on(uart)}$	The time from power-on issue to UART port ready	-	8	-	s
$T_{on(usb)}$	The time from power-on issue to USB port ready	-	9	-	s
V_{IH}	Input high level voltage on PWRKEY pin	0.7* VBAT	-	VBAT	V
V_{IL}	Input low level voltage on PWRKEY pin	0	0	0.3* VBAT	V

3.2.2 Power off

Module has the following shutdown methods:

- Power off by pulling the PWRKEY pin down to a low level.
- Power off Module by AT command 'AT+CPOF'.
- Over-voltage or under-voltage automatic power off. Use 'AT+CPMVT' to set the voltage range
- Over-temperature or under-temperature automatic power off.

It is strongly recommended that the customer use PWRKEY or 'AT+CPOF' to shut down, and then power off VBAT (especially when the module does not need to work). In addition, the customer cannot shut down VBAT by disconnecting it, which may cause flash damaged.

NOTE

1. When the temperature exceeds the range of $-30 \sim +75\text{ }^{\circ}\text{C}$, module will report warning information through AT port. When the temperature exceeds the range of $-40 \sim +85\text{ }^{\circ}\text{C}$, module will shut down automatically. For a detailed description of 'AT+ CPOF' and 'AT+ CPMVT', please refer to document [1].
2. Strongly recommended that the design has the function to turn off the power supply to the module in an abnormal state, and then turn on the power to restart the module.
3. Strongly recommended that the PWR_CTRL (DC_DC or LDO enable pin) signal be connected to the main control.

When the module is in the power on state, it can be turned off by pulling down the PWRKEY pin, power off sequence see the following figure:

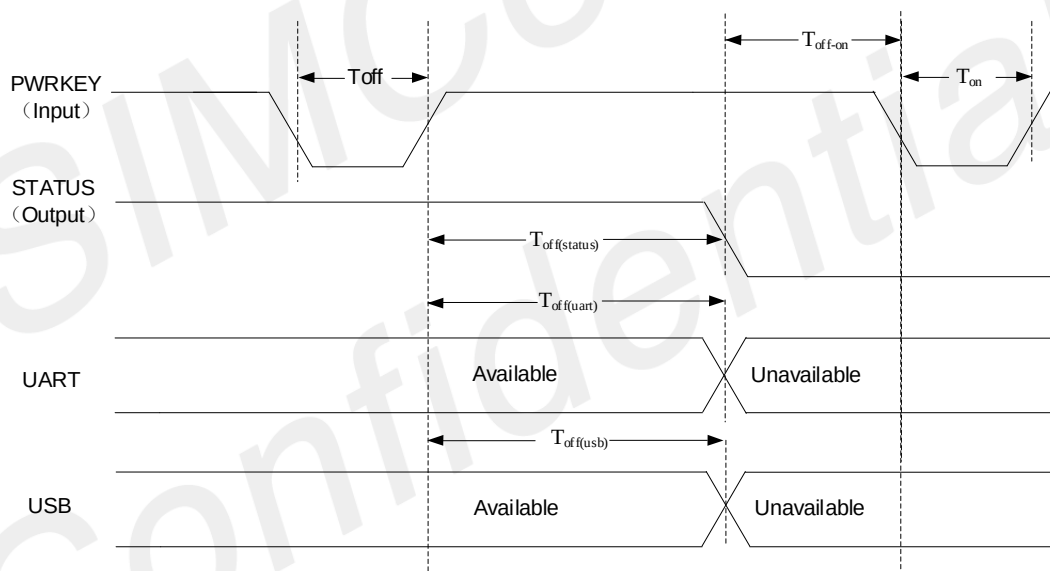


Figure 11: Power off timing sequence

Table 14: Power off sequence parameters

Symbol	Parameter	Min.	Typ.	Max.	Unit
T_{off}	Power off low level pulse width	2.5	-	-	s
$T_{off(status)}$	Power off time (according to status interface)	-	2	-	s
$T_{off(uart)}$	Power off time (according to UART interface)	-	2	-	s
$T_{off(usb)}$	Power off time (according to USB interface)	-	2	-	s
T_{off-on}	Power off - power on buffer time	2	-	-	s

3.2.3 Reset Function

Table 15: RESET interface

Pin Name	Pin No.	I/O	Description	Note
RESET	16	DI	Module reset, active low.	If unused, keep it open.

When the module is turned on, pulling down the RESET pin can reset the module, which has been pulled up to VBAT inside the module.

Recommend add TVS at the module pins to enhance anti-static capability. The recommended circuit is showed as follows:

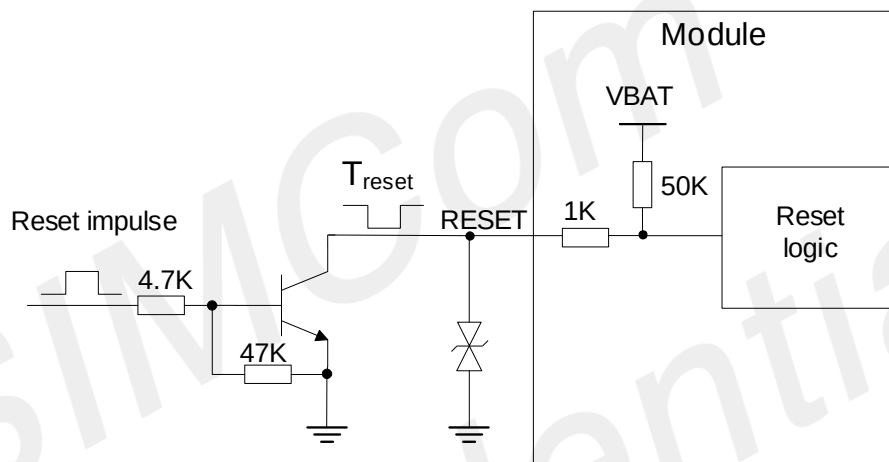


Figure 12: Reference reset circuit

Table 16: RESET pin electronic characteristic

Symbol	Description	Min.	Typ.	Max.	Unit
T_{reset}	The active low level time impulse on RESET pin to reset module	2	2.5	-	S
V_{IH}	Input high level voltage	0.7* VBAT	-	VBAT	V
V_{IL}	Input low level voltage	0	0	0.3* VBAT	V

NOTE

1. It is recommended to use the RESET pin only in emergency situations, such as when the module is not responding. The recommended RESET time is 2.5 seconds.
2. Do not connect more than 100nF capacitors in parallel to the RESET. Otherwise, the module will start up automatically if it detects a low-level during power-on. At the same time, pay attention to the TVS junction capacitance, if the junction capacitance is too large, there will be the same risk.

3.3 UART Interface

Module provides three serial ports, the main communication serial port, one ordinary serial port, and the DEBUG serial port dedicate to printing log.

Table 17: UART baud information list

UART	Support baud rate(bps)	Default baud rate	Description
Main UART	300、600、1200、2400、4800、9600、19200、38400、57600、115200、230400、460800、921600、1842000、3686400	115200	Data transmission and AT command sending
Debug UART	115200	115200	Output some logs
Ordinary UART	9600、115200、230400、921600	115200	Default for GNSS transparent transmission

Table 18: UART interface

Pin Name	Pin No.	I/O	Description	Note
DTR	3	DI	DTE ready	If unused, keep it open.
RI	4	DO	Ring indication	
DCD	5	DO	Carrier detection	
TXD	9	DO	Data output	
RXD	10	DI	Data input	
CTS	7	DO	DTE clears sending	Connected to CTS of DTE. If unused, keep it open.
RTS	8	DI	DTE request sent	Connected to RTS of DTE. If unused, keep it open.
UART3_RXD	49	DI	UART3 Data reception	If unused, keep it open.
UART3_TXD	50	DO	UART3 Data sending	
DEBUG_RXD	22	DI	Log input	If unused, keep it open.
DEBUG_TXD	23	DO	Log output	

3.3.1 UART Design Guide

When using uses full-function serial port, please refer to the following connection mode:

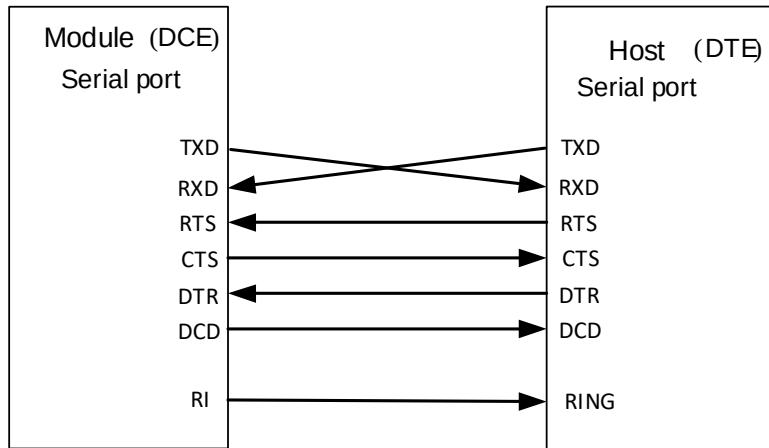


Figure 13: Serial port connection diagram (full-function mode)

When using 2-wire serial port, please refer to the following connection mode:

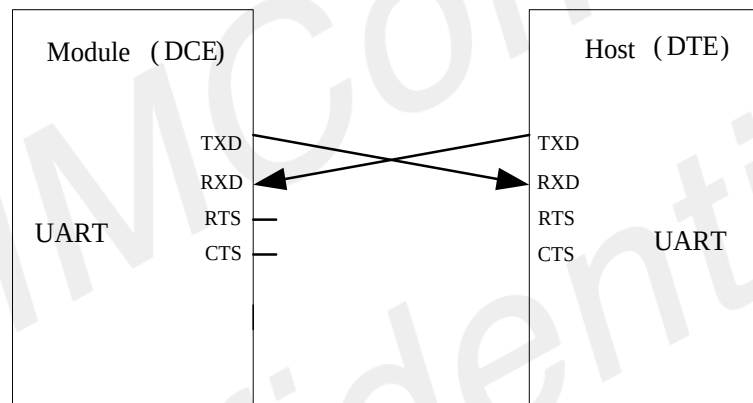


Figure 14: Serial port connection diagram (NULL mode)

The following figure shows the use of triode for level shifter circuits. The circuit with dotted line can refer to the circuit with solid line TXD and RXD, and attention shall be paid to the direction of signal. The recommended triode model is MMBT3904.

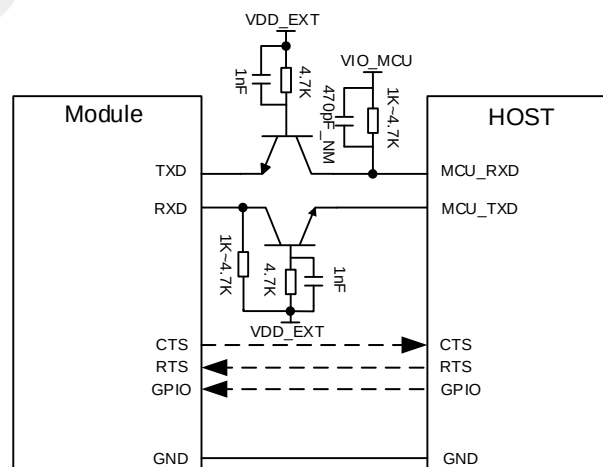


Figure 15: Triode level conversion circuit

NOTE

The parasitic capacitance of the transistor will affect the edge of the high-speed digital signal. It is not recommended to use this circuit when the signal speed is higher than 115200bps.

3.3.2 RI function

The RI pin can serve as an interrupt to wake up the host. RI usually maintains a high-level output. When receiving a short message or URC report, RI outputs a low level of 120ms (short message)/60ms (URC), and then returns to a high-level state; When receiving a phone call as the called party, RI will continue to output a 5.9s (low level)/100ms (high level) waveform until the host uses the "ATA" command to accept the call, or the calling party stops calling RI before resuming high level output after the next 100ms of high level.

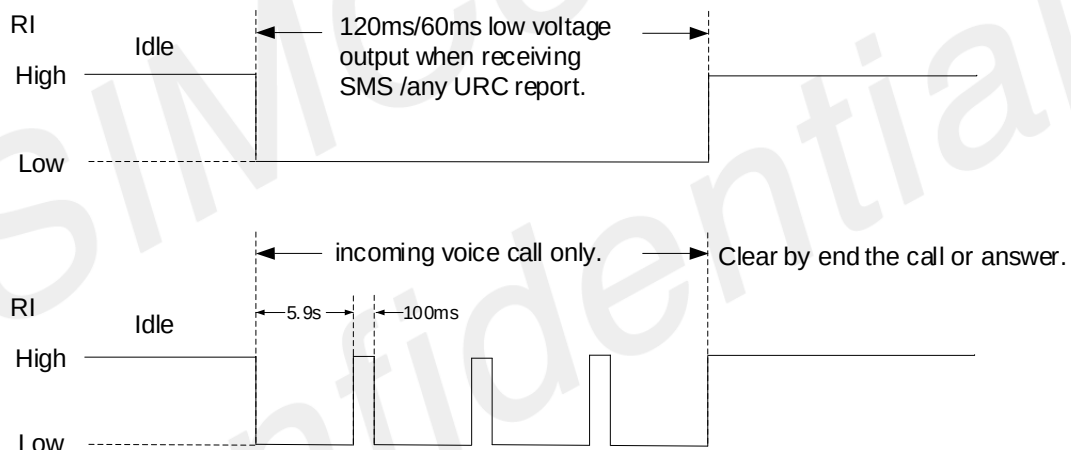


Figure 16: RI waveform (SMS and URC report)

3.3.3 DTR function

DTR can be used as the wake-up pin of the module. When the module enters sleep mode, lower the DTR to wake up the module.

After setting the AT command "AT+CSCLK=1", and then pulling up the DTR pin, Module will enter sleep mode when module is in idle mode. In sleep mode, the UART is unavailable. When module enters sleep mode, pulling down DTR can wakeup module.

After setting the AT command "AT+CSCLK=0", module will do nothing when the DTR pin is pulling up.

3.4 USB Interface

The module contains a USB interface compliant with the USB2.0 specification as a peripheral, but does not support USB charging function and does not support USB HOST mode. Supports high-speed (480Mbps) and full speed (12Mbps), and the interface can be used for AT command sending, data transmission, software debugging and upgrading.

Table 19: USB interface

Pin Name	Pin No.	I/O	Description	Note
USB_DP	27	AIO	Positive line of the differential, bi-directional USB signal.	Conforms to USB2.0 specification. 90Ω differential impedance required.
USB_DM	28	AIO	Negative line of the differential, bi-directional USB signal.	
VBUS	24	PI	Valid USB detection input.	Valid USB detection input, Active high. $V_{MIN}=3.0V$, $V_{MAX}=5.2V$

USB is the main debugging port and software upgrade interface. It is recommended that customers reserve USB test points during design. If a main control chip is connected, 0R resistors must be reserved for switching external test points during design. Note that the USB signal branch cable cannot exceed 2mm.

3.4.1 USB Reference Design

Module can be used as a USB slave device. The recommended connection circuit diagram is as follows:

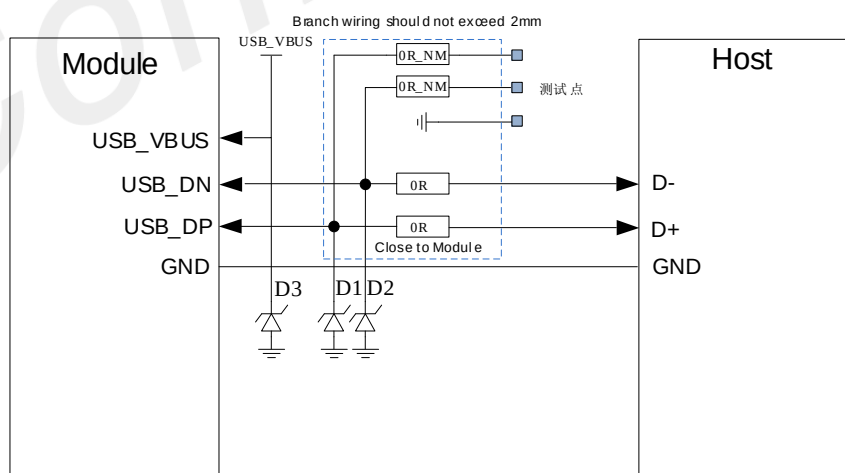


Figure 17: USB circuit diagram

In use should pay attention to the selection of D3 devices, it is recommended to choose anti-static and surge TVS devices, recommended models AZ9707-01F and WS7.0P4S1.

VBUS is the USB in-place detection pin of the module, the high level is valid, and the effective voltage is 3.0V~5.2V. It is generally connected to the VBUS pin of the external USB connector or the VBUS pin of the peripheral master USB port, and can also be connected to other power supplies such as VBAT. If the VBUS module is connected to the power supply, it is recommended to reserve a switching circuit in the middle to ensure that the power supply can be disconnected and the USB can be enumerated again

NOTE

1. The USB data cable must be strictly routed in $90\Omega \pm 10\%$ differential. The TVS devices D1 and D2 on the data line must be selected with equivalent capacitance less than 1pF. The TVS device should be placed near the USB connector or test point, recommended ESD73131C and ESD9L5.0ST5G.
2. The detection of USB2.0 speed is determined automatically by the USB protocol. The customer does not need to pull up the DP external, otherwise it may affect the device USB enumeration.

3.4.2 USB_BOOT Interface

Module provides one forced download boot interface 'USB_BOOT'.

Table 20: USB_BOOT description

Pin Name	Pin Number	I/O	Description	Remark
USB_BOOT	6	DI	Force downloads boot port	Recommend to reserve test point to facilitate debugging and upgrade. Do not pull USB_BOOT down before normal startup.

If the module upgrade fails to boot, you can force upgrade through the USB_BOOT port.

Before the module is powered on, pull the USB_BOOT pin to GND, then apply VBAT power to the module, and press RESET to enter the download mode. After entering the download mode, you need to release USB_BOOT and remove the pull-up.

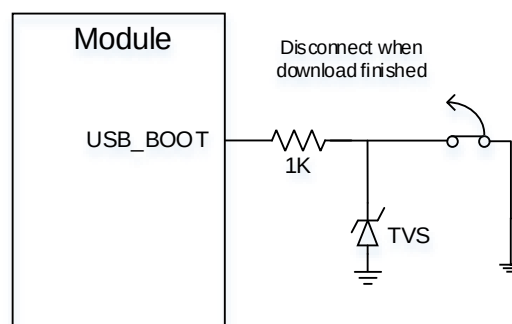


Figure 18: Reference USB_BOOT circuit

Customers will see the download port in the device manager port of the windows system.

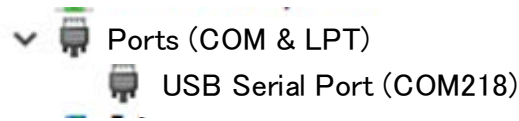


Figure 19: Force-download port

NOTE

USB_BOOT only has the function of forcing download before booting (it cannot be pulled down before normal startup). After turning on, other functions are available.

3.5 USIM Interface

Module supports both 1.8V and 3.0V USIM Cards. The interface power of the USIM card is provided by the voltage regulator inside the module, and the normal voltage value is 3V or 1.8V.

Table 21: USIM interface

Pin Name	Pin Number	I/O	Description	Remark
USIM1_DATA	31	DIO	USIM1 data	
USIM1_RST	33	DO	USIM1 reset	
USIM1_CLK	32	DO	USIM1 clock	
USIM1_VDD	30	PO	USIM1 power supply	Module automatically recognizes 1.8V or 3.0V USIM cards.
USIM1_DET	34	DI	USIM1 hot plug detection, AT instructions can be set to high/low effective, refer to documentation [25]	If unused, keep it open.
USIM2_DATA	108	DIO	USIM2 data	
USIM2_RST	111	DO	USIM2 reset	
USIM2_CLK	109	DO	USIM2 clock	
USIM2_VDD	110	PO	USIM2 power supply	Module automatically recognizes 1.8V or 3.0V USIM cards.
USIM2_DET	112	DI	USIM2 hot plug detection, AT instructions can be set to high/low effective, refer to documentation [25]	If unused, keep it open.

Table 22: USIM electronic characteristic in 1.8V mode (VUSIM=1.8V)

Symbol	Parameter	Min.	Typ.	Max.	Unit
V _{SIM}	LDO power output voltage	1.62	1.8	1.98	V
V _{IH}	High-level input voltage	0.7*USIM_VDD	-	USIM_VDD +0.4	V
V _{IL}	Low-level input voltage	-0.4	0	0.25*USIM_VDD	V
V _{OH}	High-level output voltage	USIM_VDD -0.4	-	USIM_VDD	V
V _{OL}	Low-level output voltage	0	0	0.2	V

Table 23: USIM electronic characteristic 3.0V mode (VUSIM=3V)

Symbol	Parameter	Min.	Typ.	Max.	Unit
V _{SIM}	LDO power output voltage	2.7	3	3.3	V
V _{IH}	High-level input voltage	0.7*USIM_VDD	-	USIM_VDD +0.4	V
V _{IL}	Low-level input voltage	-0.4	0	0.25*USIM_VDD	V
V _{OH}	High-level output voltage	USIM_VDD -0.45	-	USIM_VDD	V
V _{OL}	Low-level output voltage	0	0	0.3	V

3.5.1 USIM Hot Swap Function

USIM supports hot swap. You can run AT+UIMHOTSWAPON=1 to enable hot swap detection.

If the SIM card is normally open, set the module to AT+UIMHOTSWAPLEVEL=0. When the USIM_DET is lowered, the SIM card is inserted.

If the SIM card is normally closed, set the module to AT+UIMHOTSWAPLEVEL=1. When the USIM_DET is stretched high, the SIM card is inserted.

For more details on USIM hot swap AT operations, please refer to document [25]. If you do not use the USIM card hot swap function, you can keep the USIM_DET pin disconnected.

3.5.2 USIM Application Guide

The USIM interface is prone to external static interference, and suitable TVS devices must be placed near the external SIM card slot for static protection. The recommended TVS model is ESD9L5.0ST5G.

The USIM reference circuit is shown in the following figure:

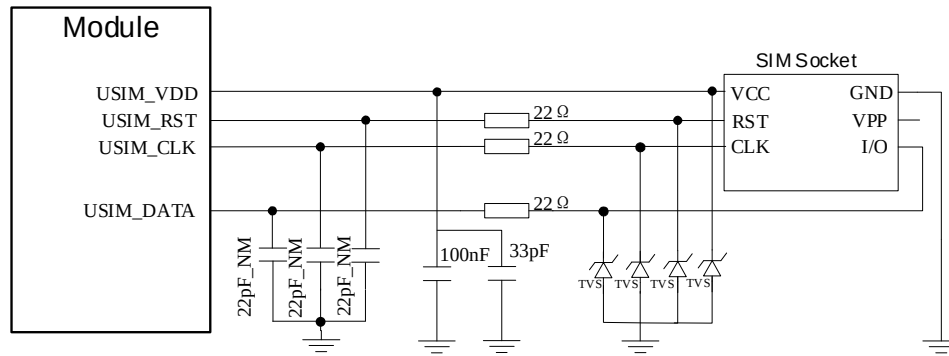


Figure 20: SIM interface reference circuit

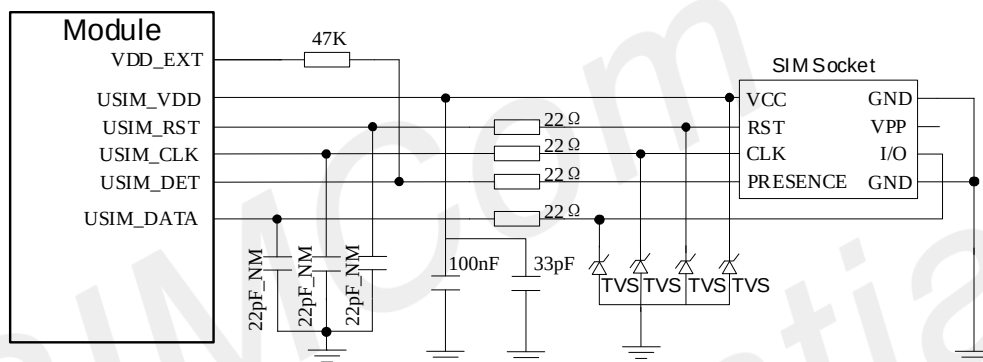


Figure 21: SIM interface reference circuit (8PIN)

NOTE

1. USIM1_DATA has been pulled up with a 4.7KΩ resistor to USIM1_VDD in module. A 100nF capacitor on USIM_VDD is used to reduce interference.
2. USIM2_DATA has no pull resistor, need to add 4.7KΩ resistor pulled up to USIM2_VDD externally.

The USIM card circuit is relatively susceptible to interference, causing situations such as not recognizing or dropping cards. Therefore, please follow the following principles when designing:

- During the PCB layout stage, it is important to keep the USIM card holder away from the main antenna.
- The USIM card should be routed as far away as possible from RF cables, VBAT, and high-speed signal cables, and the USIM card should not be too long.
- The GND of the USIM card holder should maintain good connectivity with the GND of the module, so that the GND of the two is equipotential.
- To prevent USIM_CLK interferes with other signals, it is recommended to use USIM_CLK performs separate package protection processing.
- Suggest using a 100nF capacitor on the USIM_VDD signal line near the USIM card holder.
- Place TVS near the USIM card holder, and the parasitic capacitance of the TVS should not exceed 33pF,

such as ESD9L5.0ST5G.

- Serializing a 22 Ω resistor between the USIM card holder and the module can enhance ESD protection performance.
- To ensure the most smooth wiring, it is recommended to use a single channel TVS, placed near each pin of the card holder.
- USIM_CLK signal is very important, customers should ensure the rise and fall time of the USIM_CLK signal is less than 40ns, otherwise abnormal card recognition may occur.

NOTE

If the customer is designing a vehicle product, please choose the USIM booth with better reliability.

3.6 Analog Audio Interface

3.6.1 Analog Audio Input Interface

Module provide 1 channel of analog audio MIC input interface; Voice calls can be made using an external controller.

Table 24: Analog audio MIC interface

Pin Name	Pin Number	I/O	Description	Remark
MIC_P	42	AI	Microphone input channel (+)	If unused, keep it open.
MIC_N	43	AI	Microphone input channel (-)	

3.6.2 Analog Audio Output Interface

Module provide 1 channel of analog audio SPK output interface, an external Receiver/Speaker can be used for voice calls or multimedia playback.

Table 25: Analog audio output interface

Pin Name	Pin Number	I/O	Description	Remark
EAR_P	40	AO	Handset differential output channel (+)	If unused, keep it open.
EAR_N	41	AO	Handset differential output channel (-)	

3.6.3 Analog audio electrical characteristics

Module built-in audio decoder and audio front end, the main features are as follows:

ADC: 90dB SNR@20~20kHz
DAC: 95dB SNR@20~20kHz
(Class-AB): THD<-85dB@32-ohm

Table 26: Analog audio parameter

Parameter	Conditions	DR (Typ.)	THD+N (Typ.)	Max Power
ADC	RL=10K	101dBA	-96dB (@vout - 2dBv)	1.59Vp
Class-AB	Mono, 32 Ω Difference	100dBA	-90dB (0.00316%) (@20mW output)	37mW

3.6.4 Analog audio reference design

The analog audio recommendation circuit is as follows:

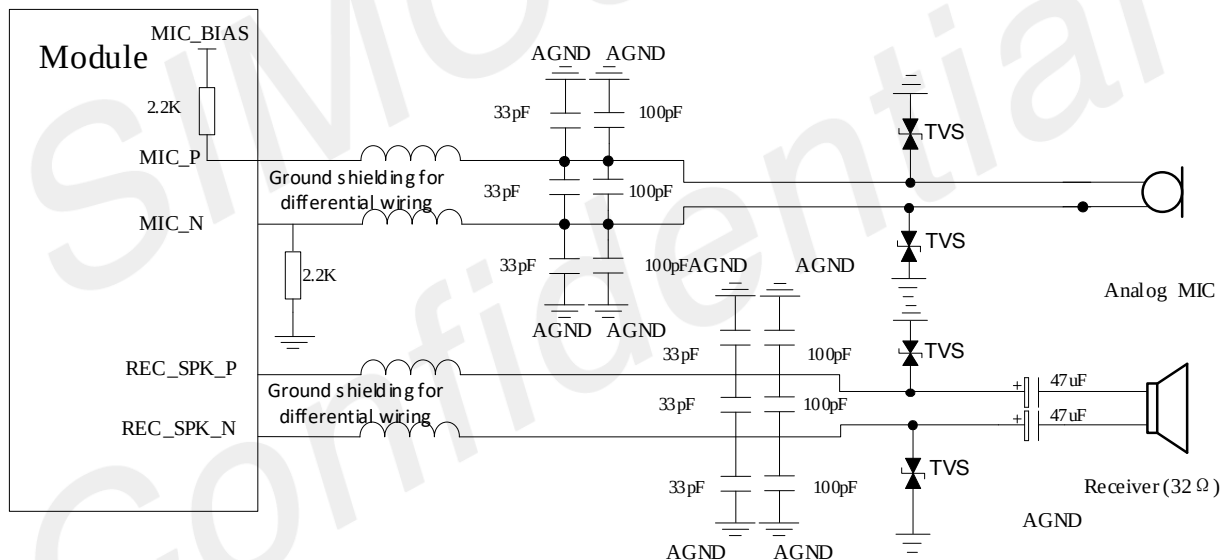


Figure 22: Analogy audio interface reference circuit

3.7 Matrix Keyboard Interface

Module provides a 4*4 matrix keyboard interface.

Table 27: Matrix keyboard PIN description

Pin Name	Pin Number	I/O	Description	Remark
MK_IN_2	47	DI	Matrix keyboard input	If unused, keep it open.

MK_IN_3	20	DI		
MK_IN_5	68	DI		
MK_IN_6	36	DI		
MK_OUT_2	44	DO	Matrix keyboard output	If unused, keep it open.
MK_OUT_3	21	DO		
MK_OUT_5	67	DO		
MK_OUT_6	35	DO		

The matrix keyboard interface recommendation circuit is as follows:

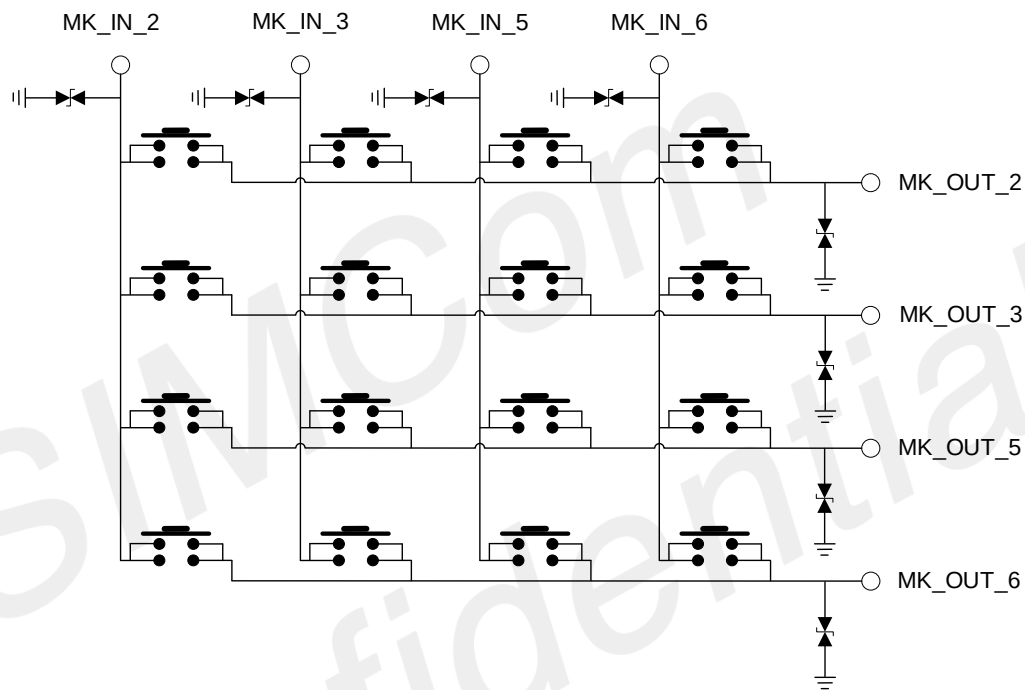


Figure 23: Matrix keyboard interface reference circuit

3.8 GPIO Interface

Table 28: Standard GPIO resources of module

Pin No.	Pin Name	AT Command Operation GPIO Number	Pin Typ.	Power Domain	Default Function	Pad Edge Wakeup
19	GPIO1	GPIO1	IO	1.8V	PU	Yes
26	GPIO2	GPIO2	IO	1.8V	PD	Yes
48	GPIO3	GPIO3	IO	1.8V	PU	No
53	GPIO4	GPIO4	IO	1.8V	PU	Yes

3.9 I2C interface

The module provides a set of I2C interface, support clock frequency up to 400Kbps, its operation voltage is 1.8V.

Table 29: I2C interface

Pin Name	Pin Number	I/O	Description	Remark
CI2C_SDA	37	OD	I2C data I/O	Need pull up to VDD_EXT with 4.7k externally. If unused, keep it open.
CI2C_SCL	38	OD	I2C clock output	

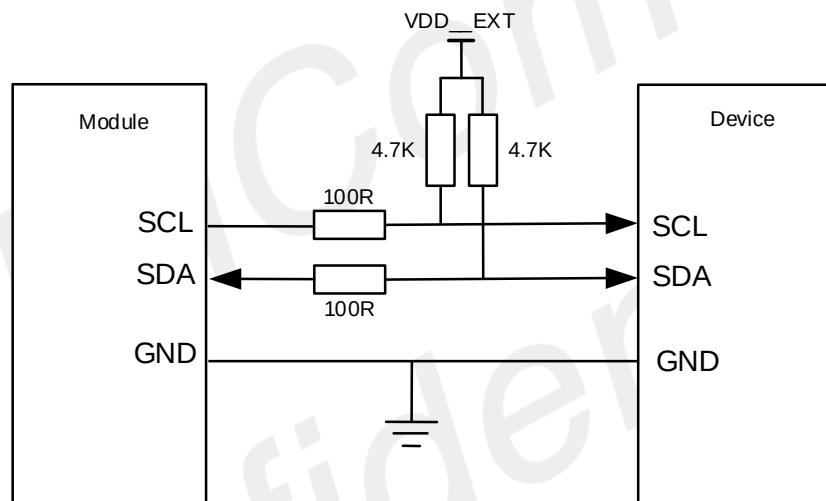


Figure 24: I2C reference circuit

NOTE

1. SCL and SDA have no pull-up resistor inside, external resistor is needed and the pulled power source must be VDD_EXT output from the module.
2. Recommend to connect 100R resistors in series near the module end of the I2C signal line to reduce overshoot on the signal line.

3.10 SPI Interface

The module provides a set of hardware SPI interfaces, supports host mode only, operates at 1.8V and has a maximum clock frequency of 52MHz.

Table 30: SPI interface

Pin Name	Pin Number	I/O	Description	Remark
SPI_CLK	11	DO	SPI bus clock output	If unused, keep it open.
SPI_CS	12	DO	SPI bus chip select	
SPI_MOSI	13	DO	SPI bus master output	
SPI_MISO	14	DI	SPI bus master input	

The recommendation circuit is as follows:

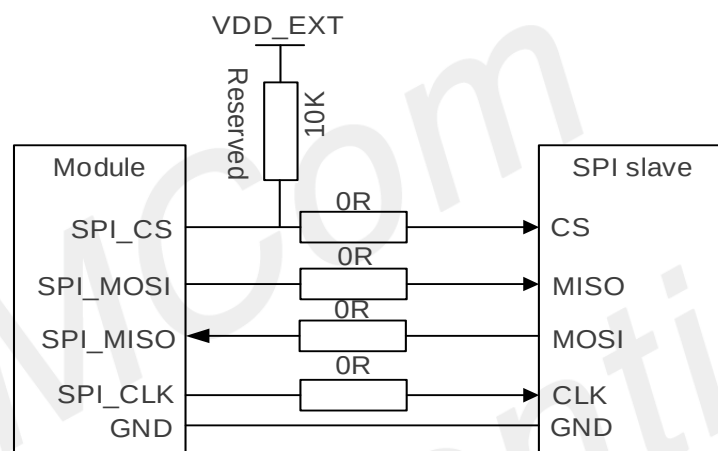


Figure 25: SPI reference circuit

3.11 Module Network Status indication

NETLIGHT can indicate the current network status of the module, usually used to drive LED lights indicating the network status.

Table 31: NETLIGHT interface

Pin Name	Pin Number	I/O	Description	Remark
NETLIGHT	52	DO	Network status indication	If unused, keep it open.

The reference circuit is shown in the following figure:

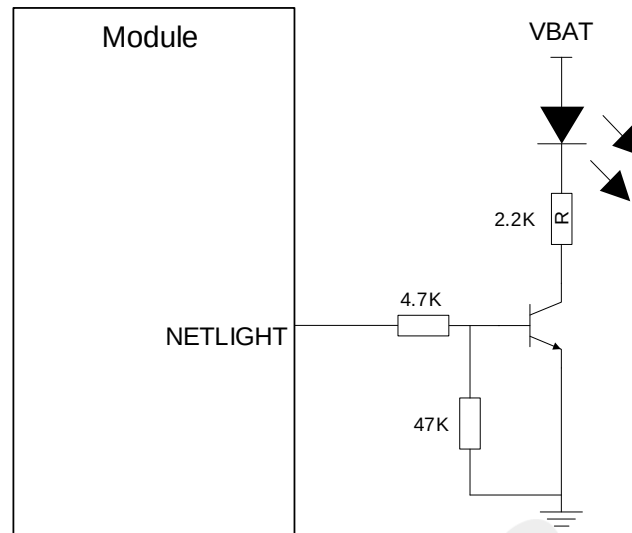


Figure 25: NETLIGHT reference circuit

NOTE

The value of the resistor named “R” depends on the VBAT and LED characteristic.

The NETLIGHT signal is used to control the LED lights that indicate the status of the network. The working status of this pin is shown in the table below.

Table 32: 2G mode NETLIGHT pin status

Net-light Pin Status	Module Status
Always On	Searching Network
200ms ON, 200ms OFF	Data Transmit
800ms ON, 800ms OFF	Registered network
OFF	Power off, or the condition satisfies AT+CSCLK=1, and the DTR is pulled up.

Table 33: LTE mode NETLIGHT pin status

Net-light Pin Status	Module Status
Always On	Searching Network
200ms ON, 200ms OFF	Data Transmit/Registered
OFF	Power off, or the condition satisfies AT+CSCLK=1, and the DTR is pulled up.

3.12 Module operation status indication

Status can indicate the operating status of the module, which is used to determine whether the module is turned on.

In earlier software versions, when the module did not have an external USB, the STATUS would briefly output a high-level pulse during the module startup process and then output a low-level signal. After the module initialization was completed, it would output a continuous high-level signal; When the module is externally connected to USB, there will be no short high-level pulses during the module startup process. Only after the module initialization is completed, the STATUS immediately outputs a continuous high-level signal.

In the latest software version, STATUS is not affected by USB and only outputs a continuous high level once the module initialization is completed.

Table 34: STATUS interface

Pin Name	Pin Number	I/O	Description	Remark
STATUS	66	DO	Module operation status indication	If unused, keep it open.

The reference circuit is shown in the following figure:

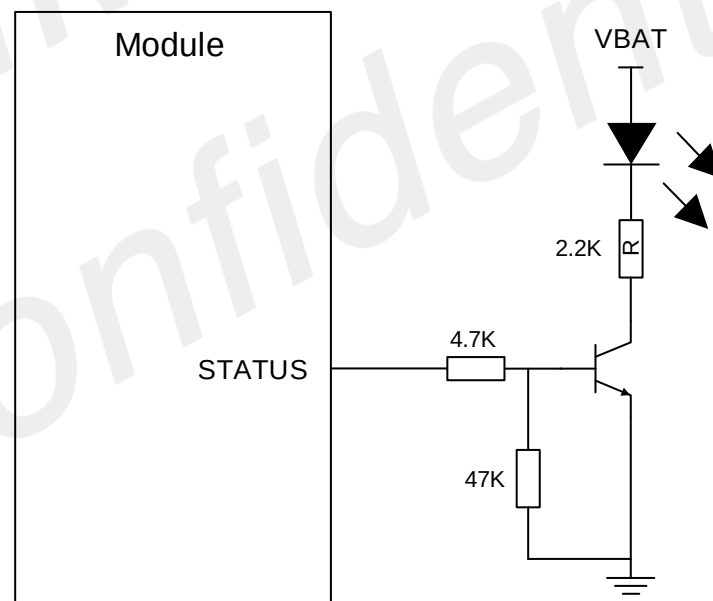


Figure 26: STATUS reference circuit

NOTE

The value of the resistor named “R” depends on the VBAT and LED characteristic.

3.13 GNSS Interface

Some models of module support GNSS function interface. GNSS provides two power supply input interfaces, one GNSS power enable control switch, one UART interface and one pulse synchronous clock signal interface, which are described in detail as follows.

Table 35: GNSS interface description

Pin Name	Pin No	I/O	Description	Note
GNSS_VBKP	116	PI	GNSS backup power input	Power supply ranges from 1.4V to 3.6V. IVBKP=1mA (Typical, @1.8V). This pin cannot be floated.
GNSS_VDD	97	PI	GNSS Vcore、VDDIO input	The power supply voltage must be no less than 1.8V and no more than 1.9V. The cable must be as short as possible, with a cable width of more than 0.3mm.
GNSS_PWRCTL	98	DI	GNSS Vcore、VDDIO power enable control	Active high. Solution 1: Use 10K resistor to Connect to GPIO, MK_IN_3 (PIN20) is recommended. Solution 2: Use 10K resistor to Connect to MCU GPIO.
GNSS_RXD	96	DI	GNSS UART RX	Solution 1:Use 1K resistor to connect UART3_TXD (PIN50) of the module in series. Solution2: Use 1K resistor to Connect to MCU UART_TX. 1.8V power domain.
GNSS_TXD	95	DO	GNSS UART TX	1.8V power domain. Solution 1 : Use 1K resistor (PIN49) of the module in series. Solution2: Use 1K resistor to Connect to MCU UART_RX
1PPS	100	DO	GNSS pulse synchronous clock signal	Second pulse signal can be used for accurate timing.
GNSS_ANT	90	AI	GNSS antenna	Antenna design reference

GNSS recommended reference design solution 1:

Module itself provides power, power enable and UART transmission to GNSS, the recommended reference design as follow:

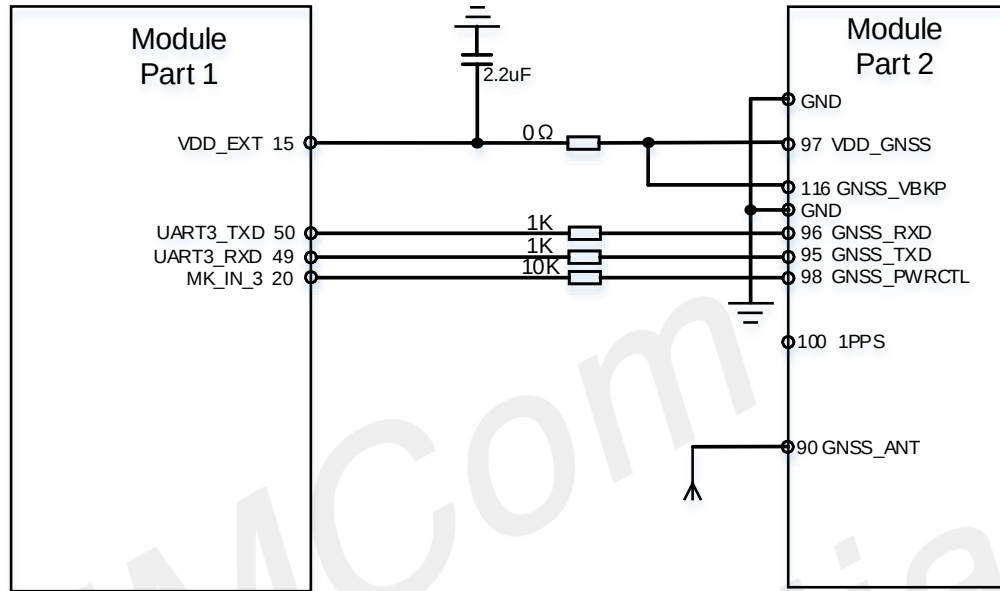


Figure 27: GNSS reference design (Non-standalone GNSS solution)

GNSS recommended reference design solution 2:

The external MCU provides power, power enable and UART transmission to GNSS, this solution is used for scenarios where GNSS can work standalone without the module powering up. The recommended reference design as follow:

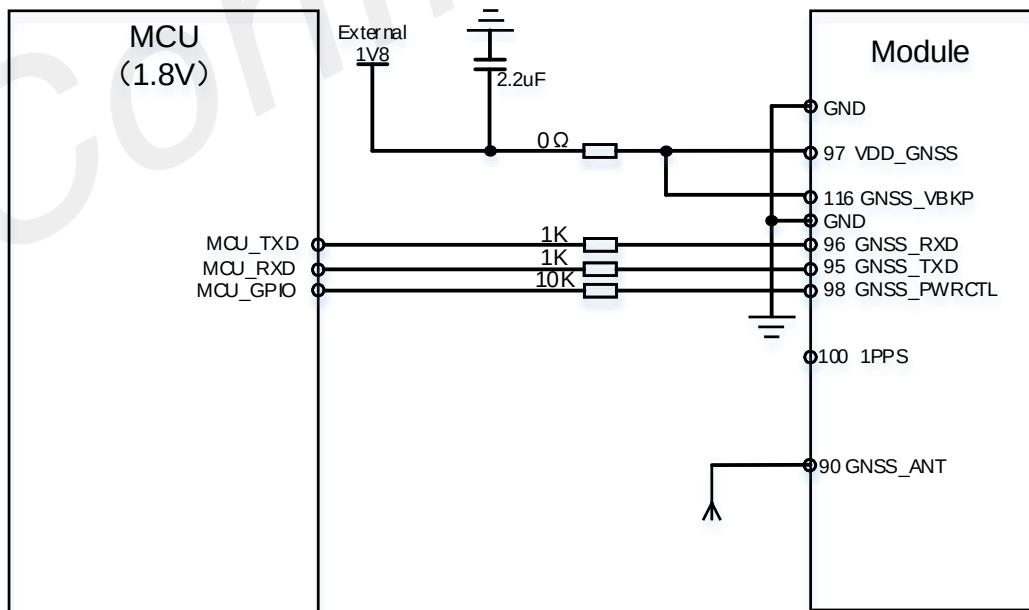


Figure 28: GNSS reference design (standalone GNSS solution)

NOTE

1. Please series in 1K resistors for serial communication lines with GNSS to prevent leakage current to the serial ports of GNSS chip.
2. The standalone GNSS reference design is only applicable to 1.8V power domain MCU. If the MCU is not 1.8V power domain, a level shift circuit should be added.
3. GNSS main power input VDD_GNSS has high requirements on power supply. The PCB cable should be as short as possible, and the cable width should be at least 0.3mm.
4. Under certain circumstances, it may be necessary to send dynamic loading code to GNSS chip through serial port. If dynamic loading is required, wiring solution 1 (non-independent GNSS solution) is recommended. If wiring scheme 2 (independent GNSS scheme) is selected, the dynamic loading process needs to be implemented by the customer's MCU.
5. GNSS_VBKP needs external power supply through VDD_EXT. This pin cannot be floated or grounded.
6. The principle of fast startup of AP-FLASH software is that after the module is successfully located, GNSS will download the location ephemeris data to the internal FLASH of the module before the module shutdown process is complete, and GNSS will download the relevant ephemeris data of the last location before the next power-on, so as to achieve fast positioning. Refer to the document [1] for detailed usage.
7. Increase the series 10K resistance of the GNSS PWRCTL line.

3.14 LCD Interface

Module provides a set of LCD interface, which only supports LCD module of 1 data line. The LCD interface of the module does not have specified LCD_TE signal pin. If necessary, you can choose GPIO to simulate the use of LCD_TE signal. It is recommended to use module pin 44 (MK_OUT_2) as the LCD_TE signal. It is recommended to reserve decoupling capacitor on the power supply for LCD, and reserve 0 Ω resistor in series for debugging. At the same time, 0 Ω in series is reserved on the data line to facilitate the adjustment of signal quality and prevent signal reflection, overshoot.

Table 36: LCD interface description

Pin Name	Pin No	I/O	Description	Note
LCD_BL_PWM	101	DO	LCD backlight PWM signal	If unused, keep it open.
LCD_SPI_CLK	102	DO	SPI clock	
LCD_SPI_TXD	103	DIO	SPI data (Bidirectional)	
LCD_SPI_RXD	104	DI	SPI data	
LCD_SPI_CS	105	DO	SPI CS	
LCD_RST	106	DO	LCD reset	
LCD_DCX	107	DO	LCD command/parameter selection	

The recommended reference design of LCD as follow:

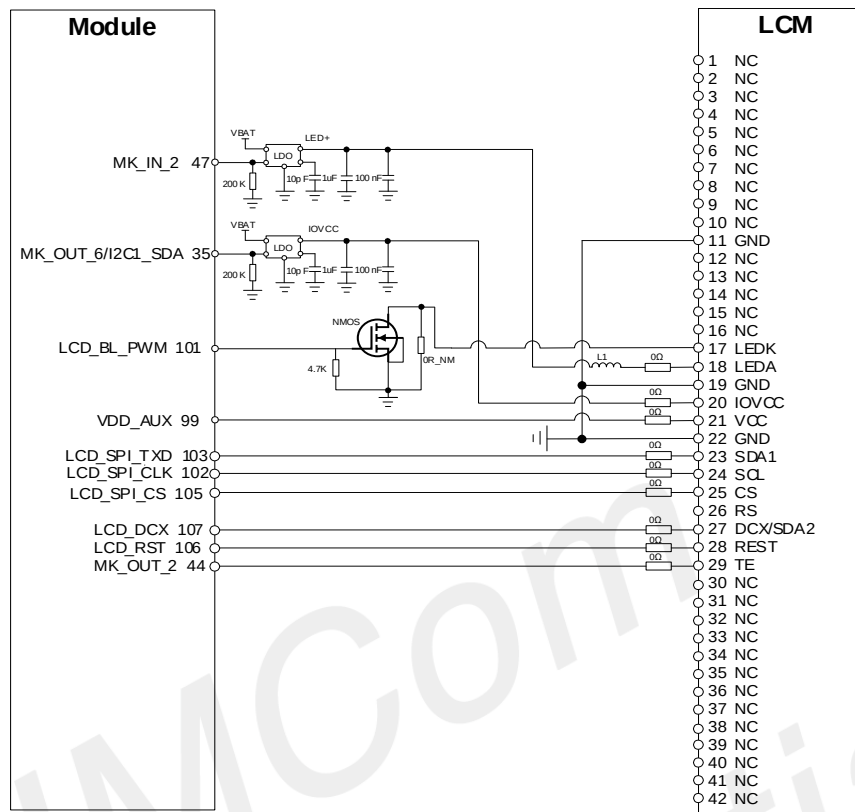


Figure 29: LCD reference design

3.15 Camera Interface

Module only supports camera interface, supports up to 0.3MP pixel encoding, does not support video mode.

Table 37: Camera interface description

Pin Name	Pin No	I/O	Description	Note
CAM_I2C_SDA	117	OD	CAM I2C data	If unused, keep it open.
CAM_I2C_SCL	118	OD	CAM I2C clock	
CAM_PWDN	119	DO	CAM power down	
CAM_RST	120	DO	CAM reset	
CAM_MCLK	121	DO	CAM main clock	
CAM_SPI_D0	122	DIO	CAM SPI DATA 0	
CAM_SPI_D1	123	DIO	CAM SPI DATA 1	
CAM_SPI_CLK	124	DO	CAM SPI clock	

The recommended reference design of camera as follow:

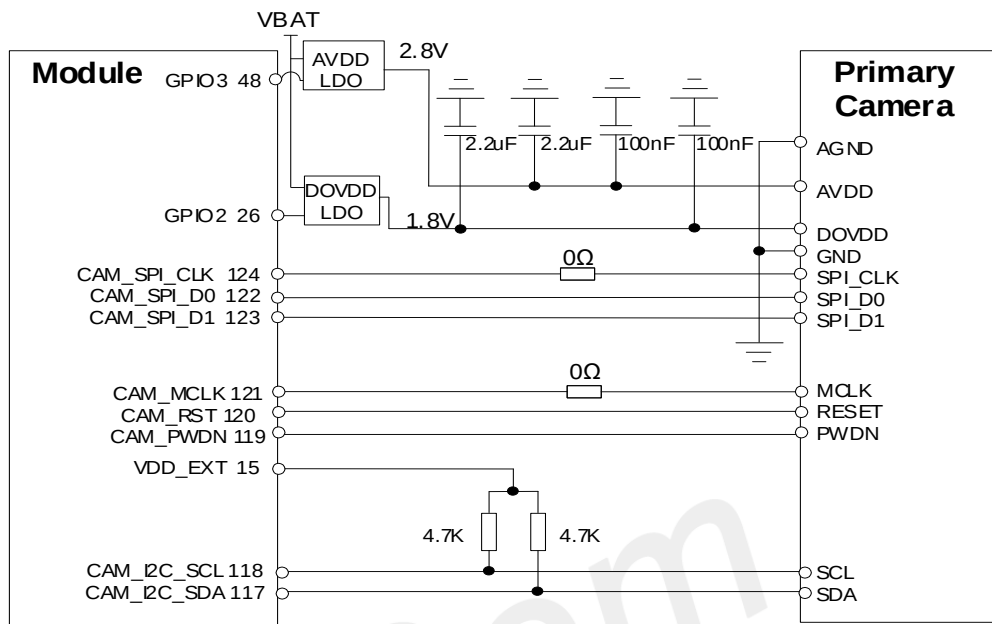


Figure 30: Camera reference design

3.16 Bluetooth Interface

Module has integrated Bluetooth function inside, and only one BT antenna is left on the module interface. Support BT5.0 protocol specification, compatible with BLE low power mode and traditional BT mode; It only supports Bluetooth data transmission, Bluetooth voice is not supported.

3.17 Another Interface

3.17.1 ADC

Module have 1 general ADC and 1 dedicated VBAT_ADC pins.

ADC input voltage range is 0~1.8V, it is recommended to directly connect to the analog interface for analog-to-digital conversion, do not perform external voltage division, if must perform external voltage division, its voltage division resistor resistance value should not be too large, it is recommended not to exceed 10K. At the same time, the amplification ratio of the partial voltage resistor should not be too large, and it is recommended not to exceed 30 times. Otherwise, the ADC read value may be offset.

The VBAT_ADC input voltage range is 0~4.2V, the default is to read the battery voltage, the hardware design VBAT_ADC peripheral must use 680K and 470K resistance for voltage division.

Table 38: ADC interface

Pin Name	Pin No	I/O	Description	Note
ADC1	25	AI	General ADC	If unused, keep it open.
VBAT_ADC	51	AI	VBAT analog to digital	

		converter interface	
--	--	---------------------	--

Table 39: General ADC electronic characteristics

Characteristics	Min.	Typ.	Max.	Unit
Resolution	—	9	-	bits
Input Range	0	-	1.8	V

Table 40: VBAT_ADC electronic characteristics

Characteristics	Min.	Typ.	Max.	Unit
Resolution	—	9	-	bits
Input Range	0	-	4.2	V

The recommended reference design of VBAT_ADC as follow:

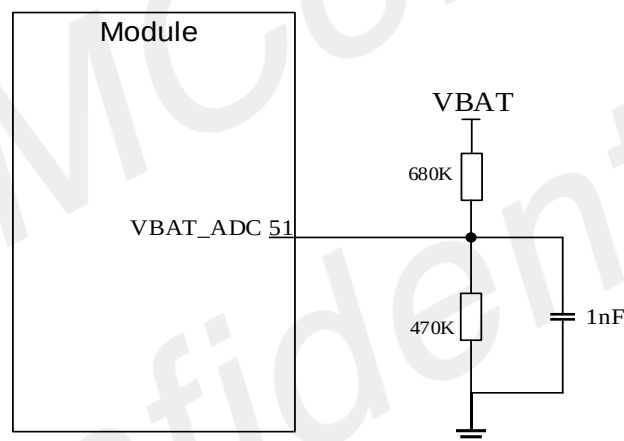


Figure 31: VBAT_ADC reference design

NOTE

1. "AT+CADC=2" can be used to read the voltage of the ADC pin(0~1.8V)
2. "AT+CBC" can be used to read the voltage of the VBAT(0~4.2V). Note that it is not the voltage value on the VBAT_ADC pin, and the design should be strictly in accordance with the VBAT_ADC reference schematic diagram.
3. The precision of the voltage divider resistance around the VBAT_ADC should not be less than 1%.
4. For more details, please refer to Document [1].

3.17.2 LDO

Module has 2 LDO outputs: VDD_EXT and VDD_AUX.

VDD_EXT is the module's system IO power supply, which can only provide a current capacity of 50mA. It

cannot be used as a high current drive source.

VDD_AUX is the module adjustable LDO power output, the default output is 3V, which can only provide a current capacity of 50mA. It can provide power for other peripherals such as active antenna and LCD. The default output voltage and voltage adjustable function depend on the actual software version.

Table 41: LDO interface

Pin Name	Pin No	I/O	Description	Note
VDD_EXT	15	PO	System IO power supply	If unused, keep it open.
VDD_AUX	99	PO	Internal LDO power supply	

Table 42: VDD_EXT electrical characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
V _{EXT}	Output voltage	-	1.8	-	V
I _{EXT}	Output current	-	-	50	mA

The recommended reference design of VDD_EXT as follow:

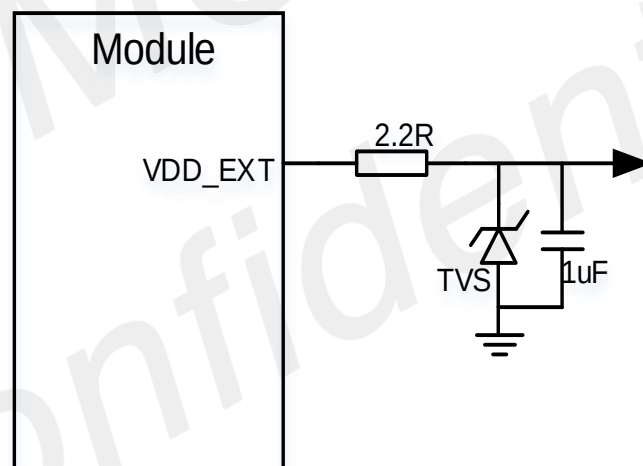


Figure 32: VDD_EXT reference design

NOTE

- VDD_EXT is the system power supply. If the damage will affect the system startup, it is recommended that customers add TVS protection. The recommended model is shown in the table below.
- It is recommended that a 2.2R resistor be connected in series near the module VDD_EXT pin for surge absorption. Since the resistor will produce a voltage drop, if the current flowing on the VDD_EXT is large, the resistance value should be appropriately reduced.

Table 43: VDD_EXT recommended TVS model list

No.	Manufacturer	Part Number	V_{RWM}	$V_C(max)$	$P_{pp}(max)$	$C_J(max)$	Package
1	WILL	ESD9X5VU	5V	8V	NA	0.9pF	WBFBP-02C
2	LRC	LESD9L5.0T5G	5V	9.8V	NA	0.9pF	SOD-923
3	ON	ESD9L5.0ST5G	5V	9.8V	NA	0.9pF	SOD-923

Table 44: VDD_AUX electrical characteristics

Symbol	Description	Min.	Typ.	Max.	Unit
V_{AUX}	Output voltage	2.5	3.0	3.0	V
I_{AUX}	Output current	-	-	50	mA

NOTE

1. VDD_AUX is the adjustable voltage output. The default output is 3.00V. The adjustable voltage values are as follows: 2.50V, 2.60V, 2.70V, 2.80V, 2.90V, 3.0V. The default output voltage and voltage adjustable function depend on the actual software version.
2. VDD_AUX cannot be turned off. Pay attention to the usage scenario.
3. Please refer to the document [1] for voltage output regulation method.

4 RF Specifications

4.1 GSM/LTE Specifications

Table 45: Conducted transmission power

Frequency	Power	Min.
GSM850	33dBm $\pm$ 2dB	5dBm $\pm$ 5dB
EGSM900	33dBm $\pm$ 2dB	5dBm $\pm$ 5dB
DCS1800	30dBm $\pm$ 2dB	0dBm $\pm$ 5dB
PCS1900	30dBm $\pm$ 2dB	0dBm $\pm$ 5dB
GSM850 (8-PSK)	27dBm $\pm$ 3dB	5dBm $\pm$ 5dB
EGSM900 (8-PSK)	27dBm $\pm$ 3dB	5dBm $\pm$ 5dB
DCS1800 (8-PSK)	26dBm +3/-4dB	0dBm $\pm$ 5dB
PCS1900 (8-PSK)	26dBm +3/-4dB	0dBm $\pm$ 5dB
LTE-FDD B1	23dBm +/-2.7dB	<-40dBm
LTE-FDD B2	23dBm +/-2.7dB	<-40dBm
LTE-FDD B3	23dBm +/-2.7dB	<-40dBm
LTE-FDD B4	23dBm +/-2.7dB	<-40dBm
LTE-FDD B5	23dBm +/-2.7dB	<-40dBm
LTE-FDD B7	23dBm +/-2.7dB	<-40dBm
LTE-FDD B8	23dBm +/-2.7dB	<-40dBm
LTE-FDD B20	23dBm +/-2.7dB	<-40dBm
LTE-FDD B28	23dBm +/-2.7dB	<-40dBm
LTE-FDD B66	23dBm +/-2.7dB	<-40dBm
LTE-TDD B34	23dBm +/-2.7dB	<-40dBm
LTE-TDD B38	23dBm +/-2.7dB	<-40dBm
LTE-TDD B39	23dBm +/-2.7dB	<-40dBm
LTE-TDD B40	23dBm +/-2.7dB	<-40dBm
LTE-TDD B41	23dBm +/-2.7dB	<-40dBm

Table 46: GSM operating bands

Frequency	Receiving	Transmission
GSM850	869~894MHz	824~849 MHz
EGSM900	925~960MHz	880~915 MHz

DCS1800	1805~1880 MHz	1710~1785 MHz
PCS1900	1930~1990 MHz	1850~1910 MHz

Table 47: E-UTRA operating bands

E-UTRA	UL Freq.	DL Freq.	Duplex Mode
1	1920~1980 MHz	2110~2170 MHz	FDD
2	1850~1910MHz	1930~1990MHz	FDD
3	1710~1785 MHz	1805~1880 MHz	FDD
4	1710~1755MHz	2110~2155MHz	FDD
5	824~849 MHz	869~894MHz	FDD
7	2500~2570MHz	2620~2690MHz	FDD
8	880~915 MHz	925~960 MHz	FDD
20	832~862MHz	791~821MHz	FDD
28	703~748MHz	758~803MHz	FDD
66	1710~1780MHz	2110~2200MHz	FDD
34	2010~2025MHz	2010~2025MHz	TDD
38	2570~2620 MHz	2570~2620 MHz	TDD
39	1880~1920 MHz	1880~1920 MHz	TDD
40	2300~2400 MHz	2300~2400 MHz	TDD
41	2535~2655 MHz	2535~2655 MHz	TDD

Table 48: Conducted receive sensitivity

Frequency	Receive sensitivity (Typical)	Receive sensitivity (Max)
GSM850	< -109dBm	3GPP
EGSM900	< -109dBm	3GPP
DCS1800	< -108dBm	3GPP
PCS1900	< -107dBm	3GPP
LTE FDD/TDD	See table 49	3GPP

Table 49: Reference sensitivity (QPSK)

E-UTRA Band	3GPP Standard						Actual 10MHz	Duplex Mode
	1.4 MHz	3MHz	5MHz	10MHz	15MHz	20MHz		
1			-97.5	-94	-92.2	-91	-96.5	FDD
2	-100.2	-97.2	-5.5	-92	-90.2	-89	-97	FDD
3	-99.2	-96.2	-94.5	-91	-89.2	-98	-97	FDD
4	-102.2	-99.2	-97.5	-94	-92.2	-91	-97.5	FDD
5	-100.7	-97.7	-95.5	-92.5			-99	FDD
7			-95.5	-92	-90.2	-89	-96	FDD
8	-99.7	-96.7	-94.5	-91.5			-99	FDD

20			-94.5	-91.5	-88.2	-87	-99	FDD
28		-97.7	-96	-93	-91.2	-88.5	-97.5	FDD
66	-101.7	-98.7	-97	-93.5	-91.7	-90.5	-97.5	FDD
34			-97.5	-94.5	-92.7	-91.5	-97.5	TDD
38			-97.5	-94.5	-92.7	-91	-97	TDD
39			-97.5	-94.5	-92.7	-91.5	-97.5	TDD
40			-97.5	-94.5	-92.7	-90	-96	TDD
41			-95.5	-92.5	-90.7	-89.5	-97.5	TDD

4.2 GSM/LTE Antenna Requirements

For better overall performance, it is recommended that the antenna design refer to the index requirements in the following table.

Table 50: GSM/LTE antenna requirements

Passive	Recommended Standard
Operating band	See table 46 and table 47
Direction	Omni Directional
Gain	> -3dBi (Avg)
Input impedance	50 ohm
Efficiency	> 50 %
Maximum input power	50W
VSWR	< 2
Isolation	>20dB
PCB insertion loss(<1GHz)	<0.5dB
PCB insertion loss(1GHz~2.2GHz)	<1dB
PCB insertion loss(2.3GHz~2.7GHz)	<1.5dB

4.3 GNSS Specifications

Table 51: GNSS operating bands

Type	Frequency
GPS	1575.42±1.023MHz
GLONASS	1597.5~1605.8MHz
BeiDou	1561.098±2.046MHz

Table 52: GNSS performance

GNSS	GPS	BeiDou	GLONASS
Tracking sensitivity	-160 dBm	-161 dBm	TBD
Cold-start sensitivity	-146.5dBm	-145dBm	TBD
Quick start time	3s		
Cold start time	<40s		
Accuracy	<2m CEP50		

NOTE

The fast startup time of GNSS is affected by the injection time of AGNSS and LBS.

4.4 GNSS Antenna Requirements

Table 53: Recommended antenna characteristics (GNSS)

Passive	Recommended Standard
Operating band	L1: 1559~1609MHZ
Direction	Hemisphere, face to sky
Input impedance	50 ohm
Maximum input power	50W
VSWR	< 2
Plan category	RHCP or Linear
Passive antenna gain	0dBi
Active antenna gain	-2dBi
Active antenna noise figure	< 1.5
Built-in antenna LNA gain	20dB(Typ.)
Total antenna gain	< 18 dB
Coaxial insertion loss	<1.5dB

4.5 Bluetooth specifications

Table 54: Bluetooth performance

Frequency

2.402GHz~2.483GHz

TX Performance (channel 0)

	DH5	2DH5	3DH5	Unit
Max TX Power (GFSK)	TBD	6.76	6.76	dBm
Min TX Power (GFSK)	TBD	-12.47	-12.47	dBm
Max TX Power (DPSK)	TBD	5.92	5.92	dBm
Min TX Power (DPSK)	TBD	-13.31	-13.31	dBm

RX Performance

	DH5	2DH5	3DH5	Unit
RX sensitivity	TBD	TBD	TBD	dBm

4.6 Antenna Reference Design

4.6.1 Passive Antenna for GSM/LTE/GNSS/Bluetooth

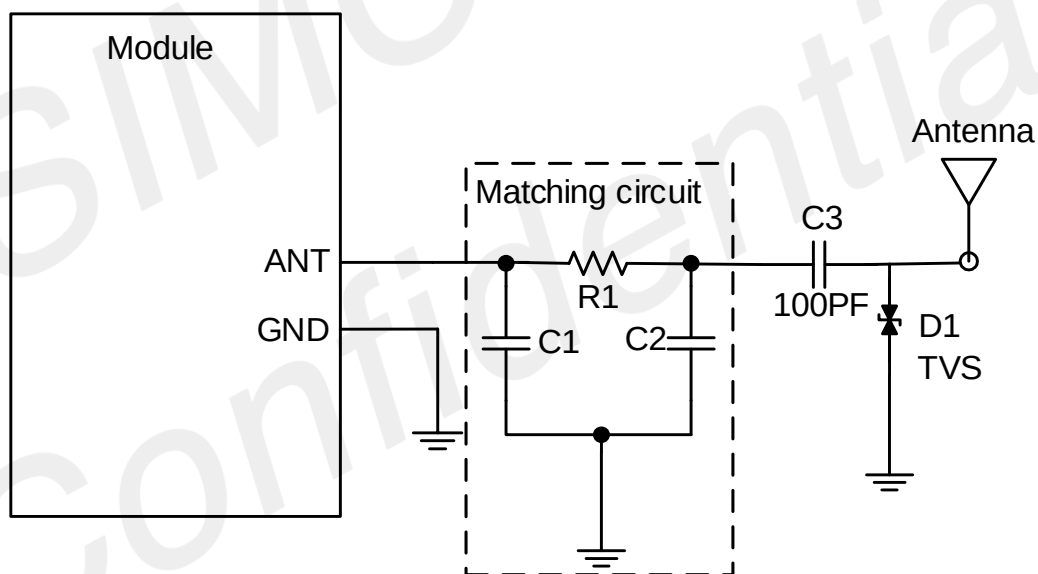


Figure 33: Passive antenna reference

The matching circuit R1 is attached to 0Ω by default, and C1 and C2 are reserved by default. The specific selection values are determined by antenna optimization and are usually provided by the antenna manufacturer.

To avoid electrostatic damage to components inside the module, C3 is recommended to install a 100pF capacitor to enhance ESD protection. D1 is recommended to install a suitable bidirectional TVS device close to the antenna. In particular, pay attention to the impact of TVS junction capacitance on RF signals.

Recommended TVS models are as follows:

Table 55: RF antenna TVS part number list

No.	Manufacturer	Part Number	V_{RWM}	$V_C(max)$	$P_{pp}(max)$	$C_J(Typ.)$	Package
1	BILLSEMI	BLE5V0CR05UB	5V	40V	NA	0.05pF	DFN1006-2

Table 56: GNSS antenna TVS part number list

No.	Manufacturer	Part Number	V_{RWM}	$V_C(max)$	$P_{pp}(max)$	$C_J(max)$	Package
1	WAYON	WE05DGCF-B	5V	23V	70W	0.3pF	DFN1006-2

4.6.2 Active Antenna for GNSS

The module VDD_AUX can be used as the power supply of GNSS active antenna, and the output voltage is controlled by AT+CVAUXV. For example, by AT+CVAUXV=2800, set the output voltage to 2.8V. AT command set once takes effect, this voltage output cannot be turned off.

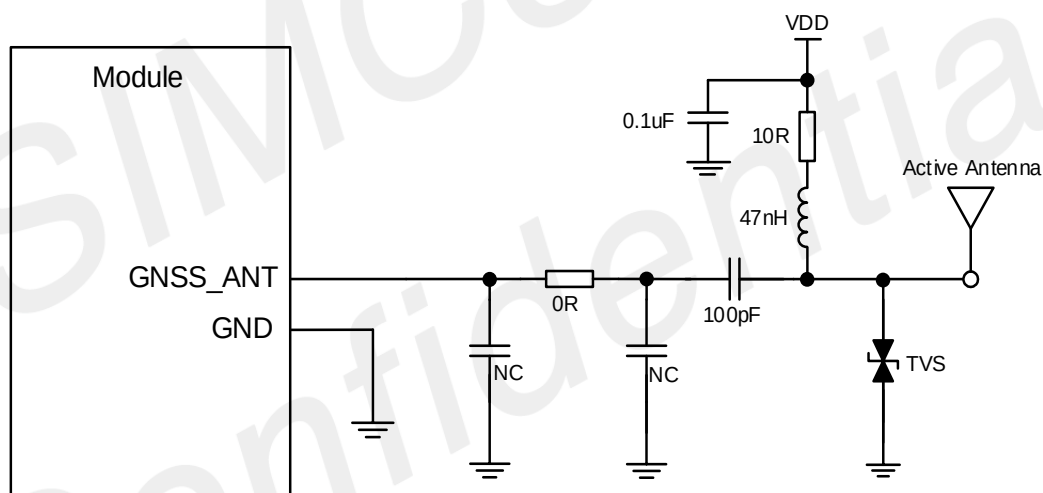


Figure 34: Active antenna reference

NOTE

1. When an active antenna is designed, the 100pF straight capacitor must be connected in series.
2. There may be parallel inductors at the internal antenna of the module, exhibiting a similar short circuit phenomenon, which is normal. If you have any doubts about this, please consult SIMCom.

4.7 PCB layout

Users should pay attention to the impedance design of PCB layout from the module ANT port to the antenna connector, and the length of the PCB trace should be within 20 mm, and far away from interference signals such as power & clock.

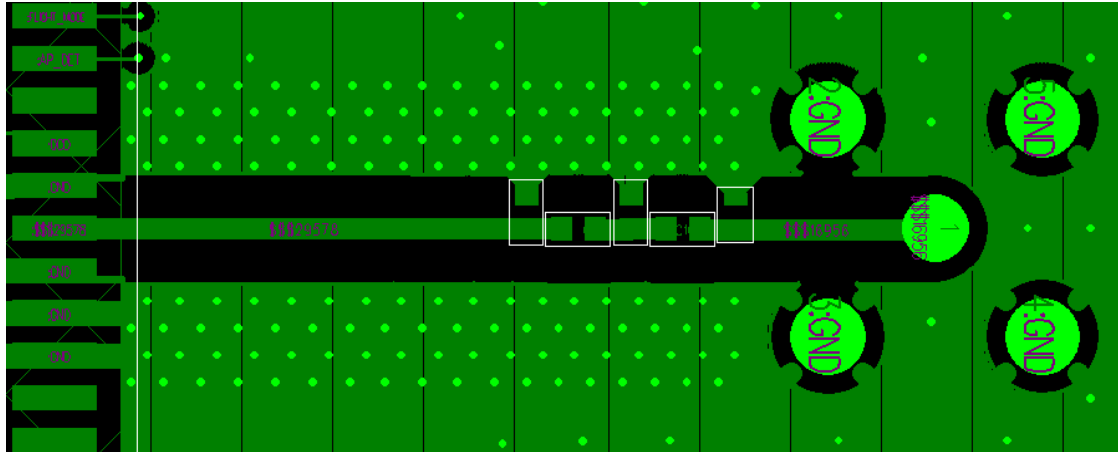


Figure 35: Reference PCB layout

5 Electrical Specifications

5.1 Absolute maximum ratings

Absolute maximum rating for digital and analog pins of module are listed in the following table, exceeding these limits may cause permanent damage to the module.

Table 57: Absolute maximum ratings

Parameter	Min.	Typ.	Max.	Unit
Voltage on VBAT	-0.5	-	4.8	V
Voltage on VBUS	-0.5	-	5.4	V
Voltage at digital pins: GPIO, UART	-0.3	-	2.0	V
Voltage at IO pins: USIM	-0.3	-	2.0	V
	-0.3	-	3.9	V
Voltage at PWRKEY, RESET	-0.3	-	4.8	V

5.2 Operating conditions

Table 58: Recommended operating ratings

Parameter	Min.	Typ.	Max.	Unit
Voltage at VBAT	3.4	3.8	4.2	V
Voltage at VBUS	3.0	5.0	5.4	V

Table 59: Operating temperature

Parameter	Min.	Typ.	Max.	Unit
Normal operation temperature	-30	25	75	°C
Extended operation temperature*	-40		85	°C
Storage temperature	-45		90	°C

NOTE

1. When working in the normal operating temperature range, the relevant performance of the module meets the requirements of 3GPP standards;
2. When the operating temperature range is extended, the module can still maintain the normal working state, with voice, short message, data transmission, emergency call and other functions; No unrecoverable failure; The radio frequency spectrum and network are basically unaffected. Only the values of individual indicators such as output power may be outside the range of the 3GPP standard. When the temperature returns to the normal operating temperature range, the indicators of the module still meet the 3GPP standard.

5.3 Operating Mode

5.3.1 Operating Mode Definition

The table below summarizes the various operating modes of module product.

Table 60: Operating mode definition

Mode		Function
Normal operation	GSM/ LTE Sleep	In this case, the current consumption of module will be reduced to the minimal level and the module can still receive paging message and SMS.
	GSM/LTE Idle	Software is active. Module is registered to the network, and the module is ready to communicate.
	GSM/ LTE Talk	Two users are connected, and in this case, the power consumption of the module is related to the network and module configuration.
	GPRS/EDGE/LTE Data transmission	There is data transmission in progress. In this case, power consumption is related to network settings (e.g., power control level); uplink/downlink data rates, etc.
Minimum functionality mode		In the case of continuous power supply, the "AT+CFUN=0" command can be used to configure the module to the minimum functional mode. In this case, both the RF part and the USIM card part are not working, but the serial port and USB can still be used, and the power consumption is lower than the normal working mode.
Flight mode		AT command 'AT+CFUN=4' or pulling down the FLIGHTMODE pin can be used to set the module to flight mode without removing the power supply. In this mode, the RF part of the

	module will not work, but the serial port and USB port are still accessible. The power consumption in this mode is lower than normal mode.
Power off	Module will go into power off mode by sending the AT command 'AT+CPOF' or pull down the PWRKEY pin, normally. In this mode the power management unit shuts down the power supply, and software is not active. The serial port and USB are is not accessible.

5.3.2 Sleep mode

In sleep mode, the current consumption of module will be reduced to the minimal level, and module can still receive paging message and SMS.

Several hardware and software conditions must be satisfied together in order to let module enter sleep mode:

- USB condition
- Software condition
- UART condition

More information about sleep mode, refer to Document [24] for more details.

5.4 Functionality mode

The module can be set to this mode by using the command "AT+CFUN=<fun>", which provides three options for setting different functions.

- AT+CFUN=0: Minimum functionality
- AT+CFUN=1: Full functionality (Default)
- AT+CFUN=4: Flight mode

After "AT+CFUN=0" is set, the module enters the minimum function mode, the RF function and USIM card function will be closed. In this case, the serial port and USB are still accessible, but RF function and USIM card will be unavailable.

After "AT+CFUN=4" is set, the module enters the flight mode, the RF function will be closed. In this case, the serial port and USB are still accessible, but RF function will be unavailable.

When module is in minimum functionality or flight mode, it can return to full functionality by the AT command "AT+CFUN=1".

For more information about the "AT+CFUN" command, see document [1].

5.5 VBAT Current Consumption

Table 61: Current consumption on VBAT pin
(VBAT=3.8V, GNSS off, BT off, without USB connection)

Minimum functional sleep state	
AT+CFUN=0, AT+CSCLK=1	Minimum functionality sleep mode Typical: 1mA
GSM sleep/idle Mode	
GSM/GPRS current	Sleep mode @BS_PA_MFRMS=2 Typical: 2.5mA Idle mode @BS_PA_MFRMS=2 Typical: 21mA
LTE sleep/idle Mode	
LTE current	Sleep mode @DRX=0.32S Typical: 2.5mA Idle mode @DRX=0.32S Typical: 21mA
GSM Talk	
GSM850	@power level #5 Typical: 220mA
EGSM900	@power level #5 Typical: 239mA
DCS1800	@power level #5 Typical: 151mA
PCS1900	@power level #0 Typical: 136mA
GPRS	
GSM850 (1 Rx,4 Tx)	@power level #5 Typical: 479mA
EGSM900 (1 Rx,4 Tx)	@power level #5 Typical: 500mA
DCS1800 (1Rx, 4 Tx)	@power level #0 Typical: 306mA
PCS1900 (1 Rx,4 Tx)	@power level #0 Typical: 300mA
GSM850 (3Rx, 2 Tx)	@power level #5 Typical: 378mA
EGSM900 (3Rx, 2 Tx)	@power level #5 Typical: 418mA
DCS1800 (3Rx, 2 Tx)	@power level #0 Typical: 253mA
PCS1900 (3Rx, 2 Tx)	@power level #0 Typical: 228mA
EDGE	
GSM850 (1 Rx,4 Tx)	@power level #8Typical: 279mA
EGSM900 (1 Rx,4 Tx)	@power level #8 Typical: 286mA
DCS1800 (1 Rx,4 Tx)	@power level #2Typical:263mA
PCS1900 (1 Rx,4 Tx)	@power level #2 Typical: 266mA
GSM850 (3Rx, 2 Tx)	@power level #8 Typical: 203mA
EGSM900 (3Rx, 2 Tx)	@power level #8Typical: 227mA
DCS1800 (3Rx, 2 Tx)	@power level #2Typical:175mA
PCS1900 (3Rx, 2 Tx)	@power level #2 Typical: 176mA
LTE Cat1	
LTE-FDD B1	@10MHz 23dBm Typical : 630mA
LTE-FDD B2	@10MHz 23dBm Typical : 590mA
LTE-FDD B3	@10MHz 23dBm Typical : 640mA
LTE-FDD B4	@10MHz 23dBm Typical : 580mA
LTE-FDD B5	@10MHz 23dBm Typical : 530mA
LTE-FDD B7	@10MHz 23dBm Typical : 630mA
LTE-FDD B8	@10MHz 23dBm Typical : 550mA
LTE-FDD B20	@10MHz 23dBm Typical : 530mA
LTE-FDD B28	@10MHz 23dBm Typical : 520mA

LTE-FDD B66	@10MHz 23dBm Typical : 550mA
LTE-TDD B34	@10MHz 23dBm Typical : 274mA
LTE-TDD B38	@10MHz 23dBm Typical : 300mA
LTE-TDD B39	@10MHz 23dBm Typical : 270mA
LTE-TDD B40	@10MHz 23dBm Typical : 270mA
LTE-TDD B41	@10MHz 23dBm Typical : 310mA

5.6 GNSS Current Consumption

Table 62: Current consumption of VBAT under GNSS working state
(VBAT=3.8V, @AT+CFUN=0, without USB connection)

Mode	Positioning system	Condition	Typical(mA)
Instrument	GPS	-130dBm/Tracking	29.93
		-145dBm/Tracking	29.06
		-130dBm/Cold boot	53.03
		-145dBm/Cold boot	52.26
		Loos of lock	53.43
Instrument	GPS+BD	-130dBm/Tracking	29.46
		-145dBm/Tracking	29.26
		-130dBm/Cold boot	52.86
		-145dBm/Cold boot	52.28
		Loos of lock	52.05
Open-sky passive antenna	GPS	Outdoor search	53.80
		Location	29.69
		Loos of lock	52.81
Open-sky passive antenna	GPS+BD	Outdoor search	53.93
		Location	29.77
		Loos of lock	53.06

5.7 ESD

Modules are electrostatic sensitive devices, therefore, users must pay attention to electrostatic protection when producing, assembling, and operating modules. Module is sensitive to ESD in the process of storage, transporting, and assembling.

The static performance parameters of the module are shown in the table below:

Table 63: The ESD performance measurement table (Temperature: 25℃, Humidity: 45%.)

Part	Contact Discharge	Air Discharge
VBAT, GND	+/-4K	+/-8K
Antenna port	+/-4K	+/-8K
USB interface	+/-4K	+/-6K
UART interface	+/-3K	+/-6K
Other PADs	+/-1K	+/-2K

NOTE

Test conditions: The module is on the SIMCom development board (the development board has the necessary ESD protection devices).

6 SMT Production Guide

6.1 Top and Bottom View of module

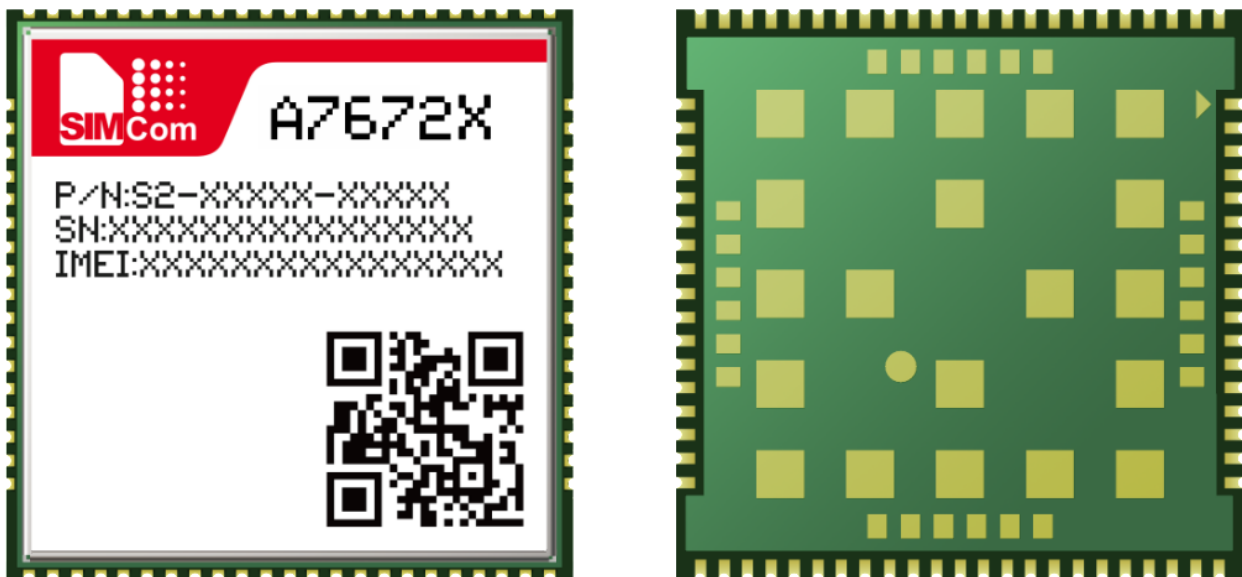


Figure 36: Top and bottom view of module

NOTE

The above is the design effect diagram of the module for reference. The actual appearance is subject to the actual product.

6.2 Label Information

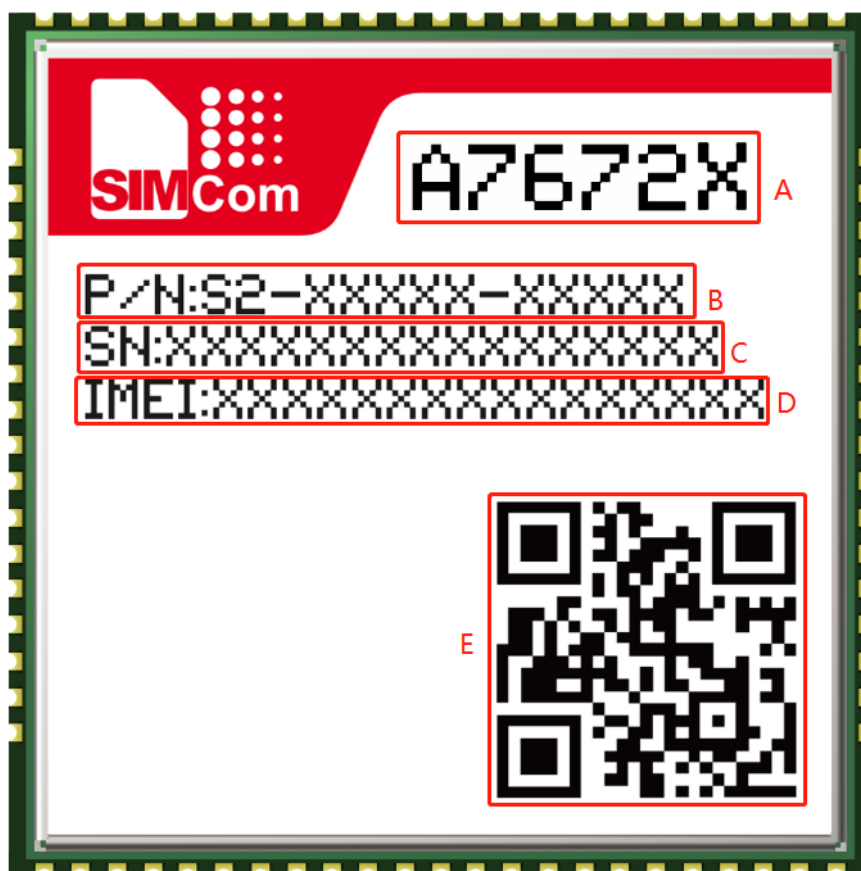


Figure 37: Label information for module

Table 64: The description of label information

No.	Description
A	Project name
B	Part number
C	Serial number
D	IMEI number
E	QR code

6.3 Storage conditions

The modules are shipped in the form of vacuum sealed aluminum foil bags, and the Vacuum packing is in accordance with the IPC/JEDEC standard in accordance with the J-STD-020C specification.

- Recommended storage conditions: temperature $23^{\circ}\text{C} \pm 5^{\circ}\text{C}$, and relative humidity 35%~60%.

- Storage life (sealed Vacuum packing): under the recommended storage conditions, the storage life is 12 months.

The module meets humidity sensitivity level 3, and the shelf life after opening is shown in Table 66.

The workshop life of products with a humidity sensitivity level of 3 is 168 hours. In an environment with a workshop temperature of $23 \pm 5^{\circ}\text{C}$ and a relative humidity of less than 60%, the module needs to undergo reflux production or other high-temperature operations within 168 hours after being unpacked. Otherwise, the module needs to be stored in an environment with a relative humidity of less than 10% (such as a moisture-proof cabinet) to maintain the dryness of the product.

Table 65: Moisture Sensitivity Level and Floor Life

Moisture Sensitivity Level (MSL)	Factory ambient $\leq +30^{\circ}\text{C}/60\% \text{ RH}$
1	Unlimited at $\leq +30^{\circ}\text{C}/85\% \text{ RH}$
2	1 year
2a	4 weeks
3	168 hours
4	72 hours
5	48 hours
5a	24 hours
6	Mandatory bake before use. After bake, it must be reflowed within the time limit specified on the label.

The following situations also require pre baking treatment:

- The storage temperature and humidity do not meet the recommended storage conditions;
- Leakage of vacuum sealed packaging bags and loose materials;
- Before module repair;
- The module was not produced or stored according to the humidity sensitive device level 3 control after being unpacked.

6.4 Roast condition

To bake, bake at $120^{\circ}\text{C} \pm 5^{\circ}\text{C}$ for 8 hours. Preferentially choose a nitrogen-filled oven.

Table 66: Roast condition

Optional baking conditions	Baking time	Notes
$120^{\circ}\text{C} \pm 5^{\circ}\text{C}$, $<5\% \text{ RH}$	8 hours	Original packaging pallet not applicable

NOTE

1. In order to prevent and reduce the occurrence of poor welding such as foaming and delamination caused by moisture, the module should be strictly controlled. It is not recommended to expose the module to the air for a long time after unpacking the vacuum package.
2. Before baking, take the module out of the package and place the bare module on the high temperature resistant appliance to avoid high temperature damage to the plastic tray or coil; The modules for secondary baking must be welded within 24 hours after baking, otherwise they need to be stored in vacuum packaging or in a drying oven.
3. Pay attention to ESD prevention when unpacking or placing a module. For example, wear ESD gloves.

6.5 Typical SMT Reflow Profile

Recommended to use lead-free process, and the recommended reflow furnace temperature curve is shown in the following figure:

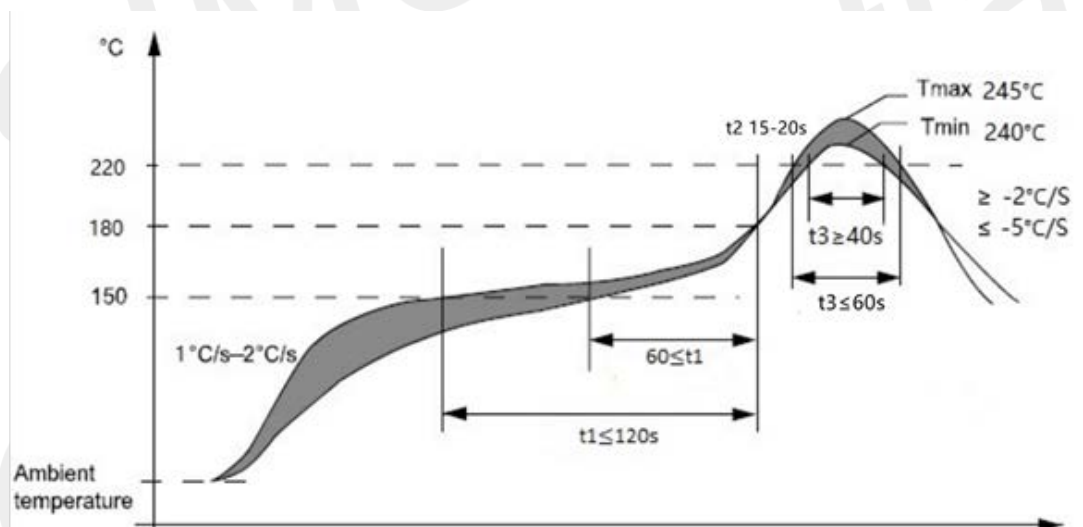


Figure 38: The ramp-soak-spoke reflow profile of module

Table 67: Explanation of temperature curve requirements for mainboard reflow soldering (lead-free)

Temperature range	Time	Critical parameter
Preheating zone (room temperature~150 °C)	NA	Temperature rise slope 1~2°C/S
T1(150~180°C)	60-120S	/
T2(180-220°C)	15-20S	/
T3(≥220°C)	40-60S	Peak temperature 240~245°C
Cooling Zone	NA	Cooling slope -2~-5°C/S
Note: The testing location includes module MCU solder joints, bottom LGA solder joints, and external LCC pin solder joints.		

Reflow times:

Maximum reflow times: 1

NOTE

1. For an introduction to module transportation, production, and other aspects, please refer to the 《Module Secondary SMT Process User Guide》.
2. Due to the complexity of SMT process, in case of uncertainty or processes not mentioned in HD documents (such as selective wave soldering, Ultrasonic welding), please confirm with SIMCom technical support before the start of SMT process.

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7 Packaging

Module supports tray packing and reel packing.

7.1 Tray packing

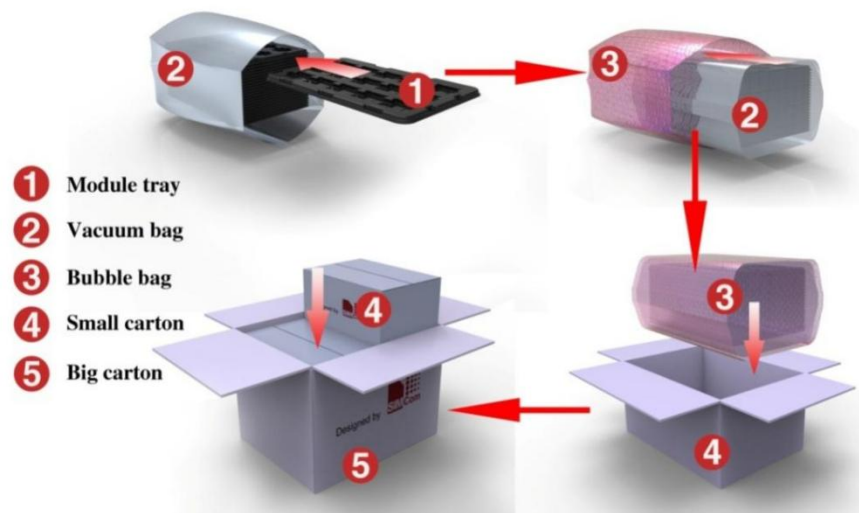


Figure 39: Packaging diagram

Module tray drawing:

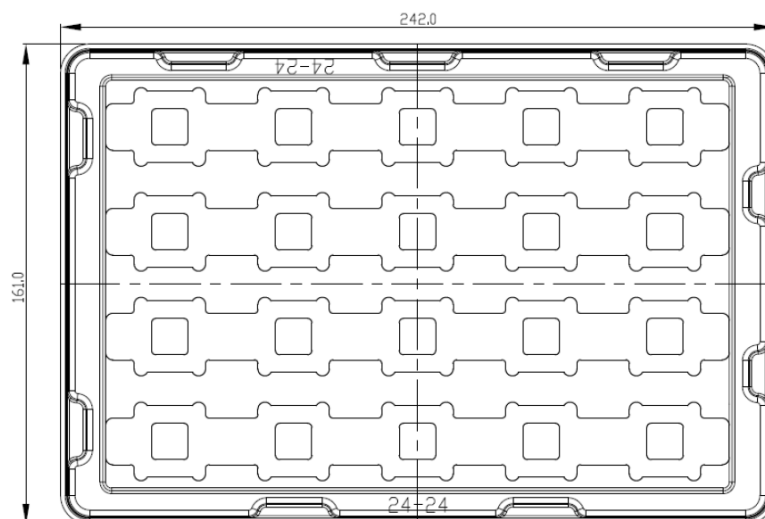


Figure 40: Tray drawing

Table 68: Tray size

Length ($\pm 3\text{mm}$)	Width ($\pm 3\text{mm}$)	Module number
242.0	161.0	20

Small carton drawing:

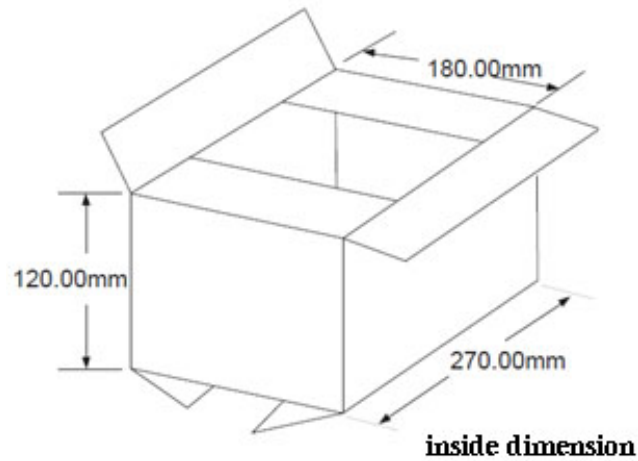


Figure 41: Small carton drawing

Table 69: Small Carton size

Length ($\pm 10\text{mm}$)	Width ($\pm 10\text{mm}$)	Height ($\pm 10\text{mm}$)	Module number
270	180	120	20*20=400

Big carton drawing (tray packaging):

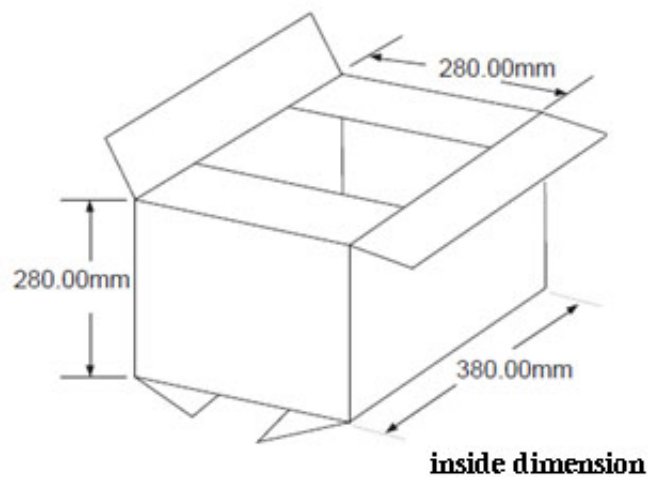


Figure 42: Big carton drawing (tray packaging)

Table 70: Big Carton size (tray packaging)

Length ($\pm 10\text{mm}$)	Width ($\pm 10\text{mm}$)	Height ($\pm 10\text{mm}$)	Module number
380	280	280	400*4=1600

7.2 Reel packing

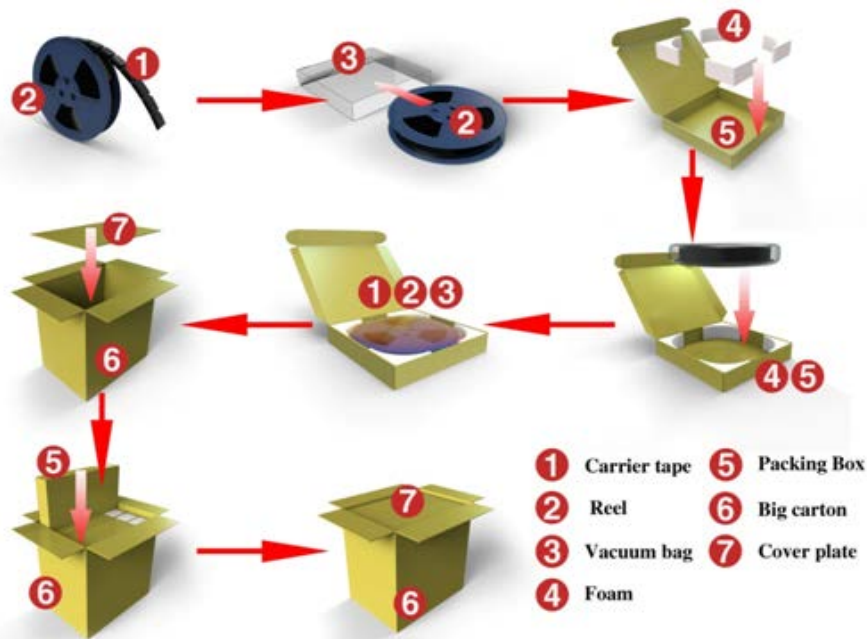


Figure 43: Reel packaging introduce

Carrier tape drawing:

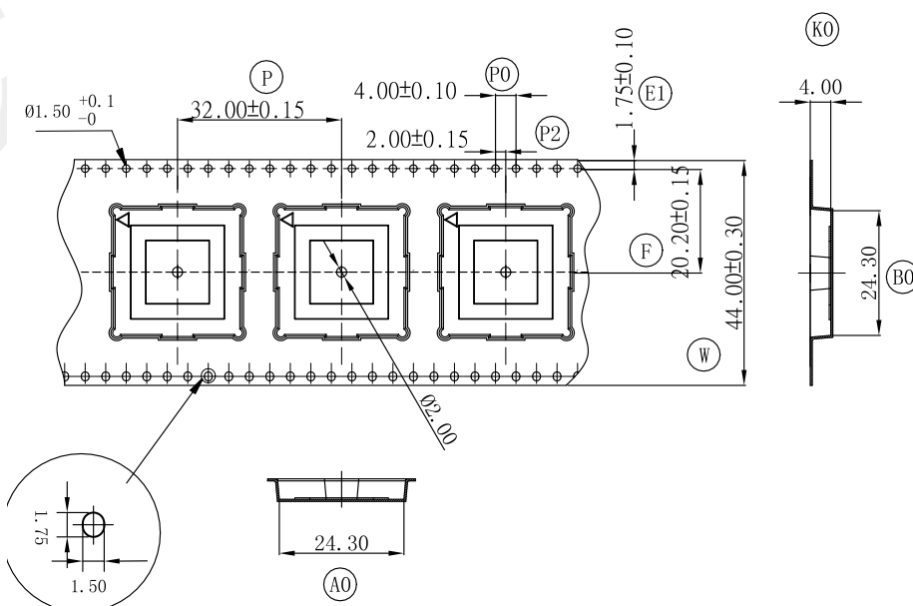


Figure 44: Carrier tape drawing introduce

Table 71: Reel size

External Diameter (mm)	Width (mm)	Inside Diameter (mm)	Module number
330.0	44.0	180.0	300

Packing box drawing:

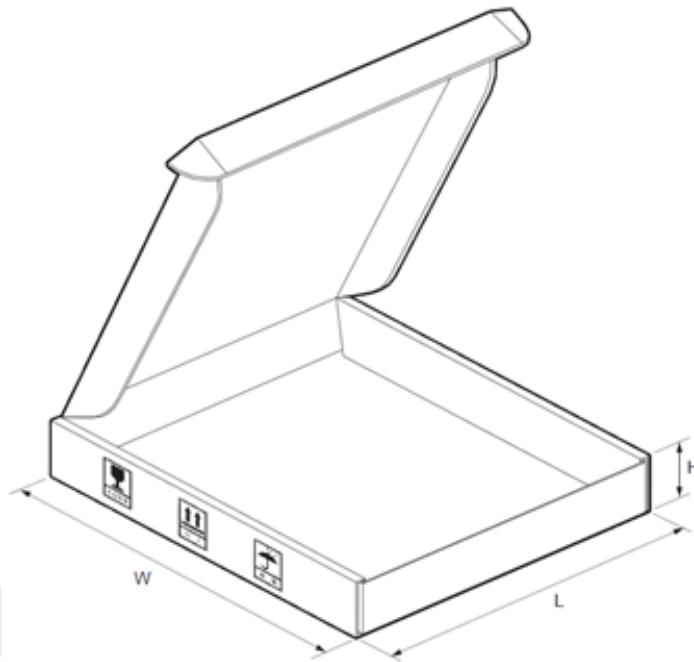


Figure 45: Reel packing box drawing introduce

Table 72: Reel packing box size

Length ($\pm 10\text{mm}$)	Width ($\pm 10\text{mm}$)	Height ($\pm 10\text{mm}$)	Module number
345	340	60	300

Big carton drawing (tray packaging):

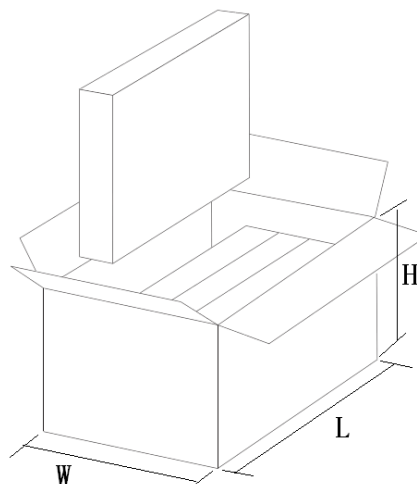


Figure 46: Big carton drawing introduce (reel packaging)

Table 73: Big carton size (reel packaging)

Length ($\pm 10\text{mm}$)	Width ($\pm 10\text{mm}$)	Height ($\pm 10\text{mm}$)	Module number
380	275	380	300*4=1200

8 Appendix

8.1 Coding Schemes and Maximum Net Data Rates over Air Interface

Table 74: Coding Schemes and Maximum Net Data Rates over Air Interface

Multislot definition (GPRS/EDGE)			
Slot Class	DL Slot Number	UL Slot Number	Active Slot Number
1	1	1	2
2	2	1	3
3	2	2	3
4	3	1	4
5	2	2	4
6	3	2	4
7	3	3	4
8	4	1	5
9	3	2	5
10	4	2	5
11	4	3	5
12	4	4	5
GPRS Coding Scheme	Max Data Rate (4 slots)		Modulation Type
CS 1 = 9.05 kb/s / time slot	36.2 kb/s		GMSK
CS 2 = 13.4 kb/s / time slot	53.6 kb/s		GMSK
CS 3 = 15.6 kb/s / time slot	62.4 kb/s		GMSK
CS 4 = 21.4 kb/s / time slot	85.6 kb/s		GMSK
EDGE Coding Scheme	Max Data Rate (4 slots)		Modulation Type
MCS 1 = 8.8 kb/s/ time slot	35.2 kb/s		GMSK
MCS 2 = 11.2 kb/s/ time slot	44.8 kb/s		GMSK
MCS 3 = 14.8 kb/s/ time slot	59.2 kb/s		GMSK
MCS 4 = 17.6 kb/s/ time slot	70.4 kb/s		GMSK
MCS 5 = 22.4 kb/s/ time slot	89.6 kb/s		8PSK
MCS 6 = 29.6 kb/s/ time slot	118.4 kb/s		8PSK
MCS 7 = 44.8 kb/s/ time slot	179.2 kb/s		8PSK
MCS 8 = 54.4 kb/s/ time slot	217.6 kb/s		8PSK
MCS 9 = 59.2 kb/s/ time slot	236.8 kb/s		8PSK

LTE-FDD Category (Downlink)	Max Data Rate (peak)	Modulation Type
Category M1	1Mbps	QPSK/16QAM
LTE-FDD Category (Uplink)	Max Data Rate (peak)	Modulation Type
Category M1	375kbps	QPSK/16QAM

8.2 Related Documents

Table 75: Related Documents

NO.	Title	Description
[1]	A76XX Series_AT Command Manual	AT Command Manual
[2]	ITU-T Draft new recommendation V.25ter	Serial asynchronous automatic dialing and control
[3]	GSM 07.07	Digital cellular telecommunications (Phase 2+); AT command set for GSM Mobile Equipment (ME)
[4]	GSM 07.10	Support GSM 07.10 multiplexing protocol
[5]	GSM 07.05	Digital cellular telecommunications (Phase 2+); Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE – DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)
[6]	GSM 11.14	Digital cellular telecommunications system (Phase 2+); Specification of the SIM Application Toolkit for the Subscriber Identity Module – Mobile Equipment (SIM – ME) interface
[7]	GSM 11.11	Digital cellular telecommunications system (Phase 2+); Specification of the Subscriber Identity Module – Mobile Equipment (SIM – ME) interface
[8]	GSM 03.38	Digital cellular telecommunications system (Phase 2+); Alphabets and language-specific information
[9]	GSM 11.10	Digital cellular telecommunications system (Phase 2) ; Mobile Station (MS) conformance specification; Part 1: Conformance specification
[10]	3GPP TS 51.010-1	Digital cellular telecommunications system (Release 5); Mobile Station (MS) conformance specification
[11]	3GPP TS 34.124	Electromagnetic Compatibility (EMC) for mobile terminals and ancillary equipment.
[12]	3GPP TS 34.121	Electromagnetic Compatibility (EMC) for mobile terminals and ancillary equipment.
[13]	3GPP TS 34.123-1	Technical Specification Group Radio Access Network; Terminal conformance specification; Radio transmission and reception (FDD)
[14]	3GPP TS 34.123-3	User Equipment (UE) conformance specification; Part 3: Abstract Test Suites.
[15]	EN 301 908-02 V2.2.1	Electromagnetic compatibility and Radio spectrum Matters (ERM); Base Stations (BS) and User Equipment (UE) for IMT-2000. Third Generation cellular networks; Part 2: Harmonized EN for IMT-2000, CDMA Direct Spread (UTRA FDD) (UE) covering essential requirements of article

		3.2 of the R&TTE Directive
[16]	EN 301 489-24 V1.2.1	Electromagnetic compatibility and Radio Spectrum Matters (ERM); Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 24: Specific conditions for IMT-2000 CDMA Direct Spread (UTRA) for Mobile and portable (UE) radio and ancillary equipment
[17]	IEC/EN60950-1(2001)	Safety of information technology equipment (2000)
[18]	3GPP TS 51.010-1	Digital cellular telecommunications system (Release 5); Mobile Station (MS) conformance specification
[19]	GCF-CC V3.23.1	Global Certification Forum - Certification Criteria
[20]	2002/95/EC	Directive of the European Parliament and of the Council of 27 January 2003 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS)
[21]	Module secondary-SMT-UGD	Module secondary SMT Guidelines
[22]	A76XX Series_UART_Application Note	This document describes how to use UART interface of SIMCom modules.
[23]	Antenna design guidelines for diversity receiver system	Antenna design guidelines for diversity receiver system
[24]	A76XX Series_SleepMode_Application Note	Sleep Mode Application Note
[25]	A76XX Series_UIM HOT SWAP_Application Note	This document introduces UIM card detection and UIM hot swap.

8.3 Terms and Abbreviations

Table 76: Terms and Abbreviations

Abbreviation	Description
ADC	Analog-to-Digital Converter
ARP	Antenna Reference Point
BER	Bit Error Rate
BD	BeiDou
BTS	Base Transceiver Station
CS	Coding Scheme
CSD	Circuit Switched Data
CTS	Clear to Send
DAC	Digital-to-Analog Converter
DSP	Digital Signal Processor
DTE	Data Terminal Equipment (typically computer, terminal, printer)
DTR	Data Terminal Ready
DTX	Discontinuous Transmission
DAM	Downloadable Application Module
DPO	Dynamic Power Optimization
EFR	Enhanced Full Rate
EGSM	Enhanced GSM
EMC	Electromagnetic Compatibility
ESD	Electrostatic Discharge
ETS	European Telecommunication Standard
FCC	Federal Communications Commission (U.S.)
FD	SIM fix dialing phonebook
FDMA	Frequency Division Multiple Access
FR	Full Rate
GMSK	Gaussian Minimum Shift Keying
GNSS	Global Navigation Satellite System
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global Standard for Mobile Communications
HR	Half Rate
I2C	Inter-Integrated Circuit
IMEI	International Mobile Equipment Identity
LTE	Long Term Evolution
MO	Mobile Originated
MS	Mobile Station (GSM engine), also referred to as TE

MT	Mobile Terminated
NMEA	National Marine Electronics Association
PAP	Password Authentication Protocol
PBCCH	Packet Switched Broadcast Control Channel
PCB	Printed Circuit Board
PCS	Personal Communication System, also referred to as GSM 1900
RF	Radio Frequency
RMS	Root Mean Square (value)
RTC	Real Time Clock
SIM	Subscriber Identification Module
SMS	Short Message Service
SMPS	Switched-mode power supply
TDMA	Time Division Multiple Access
TE	Terminal Equipment, also referred to as DTE
TX	Transmit Direction
UART	Universal Asynchronous Receiver & Transmitter
VSWR	Voltage Standing Wave Ratio
SM	SIM phonebook
NC	Not connect
EDGE	Enhanced data rates for GSM evolution
ZIF	Zero intermediate frequency
WCDMA	Wideband Code Division Multiple Access
VCTCXO	Voltage control temperature-compensated crystal oscillator
SIM	Universal subscriber identity module
UMTS	Universal mobile telecommunications system
UART	Universal asynchronous receiver transmitter
PSM	Power saving mode

Telephone book abbreviation

FD	SIM fix dialing phonebook
LD	SIM last dialing phonebook (list of numbers most recently dialed)
MC	Mobile Equipment list of unanswered MT calls (missed calls)
ON	SIM (or ME) own numbers (MSISDNs) list
RC	Mobile Equipment list of received calls
SM	SIM phonebook
NC	Not connect

8.4 Safety Caution

Pay attention to the following safety precautions when using or repairing any terminal or mobile phone containing modules. The terminal device should inform the user of the following security information. Otherwise, SIMCom will not be responsible for any consequences arising from the user's failure to follow these warnings.

Table 77: Safety Caution

Marks	Requirements
	When in a hospital or other health care facility, observe the restrictions about the use of mobiles. Switch the cellular terminal or mobile off, medical equipment may be sensitive and not operate normally due to RF energy interference.
	Switch off the cellular terminal or mobile before boarding an aircraft. Make sure it is switched off. The operation of wireless appliances in an aircraft is forbidden to prevent interference with communication systems. Forgetting to think much of these instructions may impact the flight safety, or offend local legal action, or both.
	Do not operate the cellular terminal or mobile in the presence of flammable gases or fumes. Switch off the cellular terminal when you are near petrol stations, fuel depots, chemical plants or where blasting operations are in progress. Operation of any electrical equipment in potentially explosive atmospheres can constitute a safety hazard.
	Your cellular terminal or mobile receives and transmits radio frequency energy while switched on. RF interference can occur if it is used close to TV sets, radios, computers or other electric equipment.
	Road safety comes first! Do not use a hand-held cellular terminal or mobile when driving a vehicle, unless it is securely mounted in a holder for hands free operation. Before making a call with a hand-held terminal or mobile, park the vehicle.
	GSM cellular terminals or mobiles operate over radio frequency signals and cellular networks and cannot be guaranteed to connect in all conditions, especially with a mobile fee or an invalid SIM card. While you are in this condition and need emergent help, please remember to use emergency calls. In order to make or receive calls, the cellular terminal or mobile must be switched on and in a service area with adequate cellular signal strength. Some networks do not allow for emergency call if certain network services or phone features are in use (e.g., lock functions, fixed dialing etc.). You may have to deactivate those features before you can make an emergency call. Also, some networks require that a valid SIM card be properly inserted in the cellular terminal or mobile.